

(12) **United States Patent**
Christensen

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(45) **Date of Patent:** **Sep. 2, 2025**

(54) **DUAL DRAINAGE BAG, ASSEMBLIES, AND RELATED METHODS**

33/00; B65D 33/14; B65D 75/56; B65D 75/563; B65D 75/566; A61M 1/69; A61M 1/604; A61M 2209/082; A61M 5/1417; A61M 2202/0496

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See application file for complete search history.

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

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(65) **Prior Publication Data**

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(63) Continuation of application No. 16/836,585, filed on Mar. 31, 2020, now Pat. No. 11,730,875.

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(51) **Int. Cl.**

A61M 1/00 (2006.01)

A61G 7/05 (2006.01)

Primary Examiner — Adam Marcetich

(74) *Attorney, Agent, or Firm* — Dorsey & Whitney LLP

(52) **U.S. Cl.**

CPC **A61M 1/69** (2021.05); **A61G 7/0503** (2013.01); **A61M 1/86** (2021.05); **A61M 2202/0014** (2013.01); **A61M 2205/3379** (2013.01)

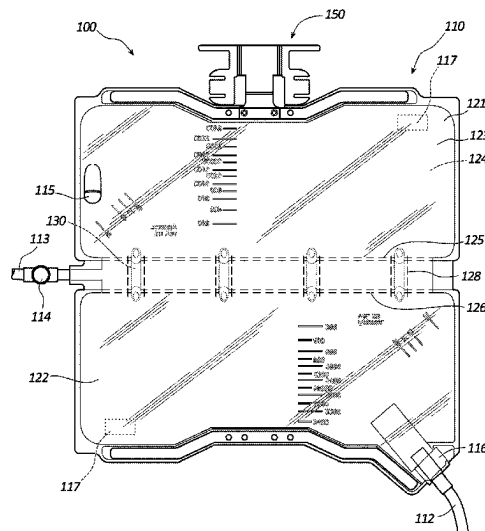
(57) **ABSTRACT**

Devices used to collect bodily fluids are disclosed. The devices may include a fluid bag and a transformable hanger for mounting the fluid bag to a support structure such as a hospital bed rail. The fluid bag can include a first fluid compartment and a second fluid compartment. The devices may further include a displaceable front cover and a back cover to obscure the bodily fluids from observation.

(58) **Field of Classification Search**

CPC A61J 1/10; A61J 1/1462; A61J 1/1475; A61J 1/2024; A61J 1/2093; A61J 1/00; A61J 1/16; A61J 1/1487; A61J 1/2034; A61G 7/0503; A61F 5/44; A61F 5/4404; B65D 1/0292; B65D 2231/001; B65D

20 Claims, 24 Drawing Sheets



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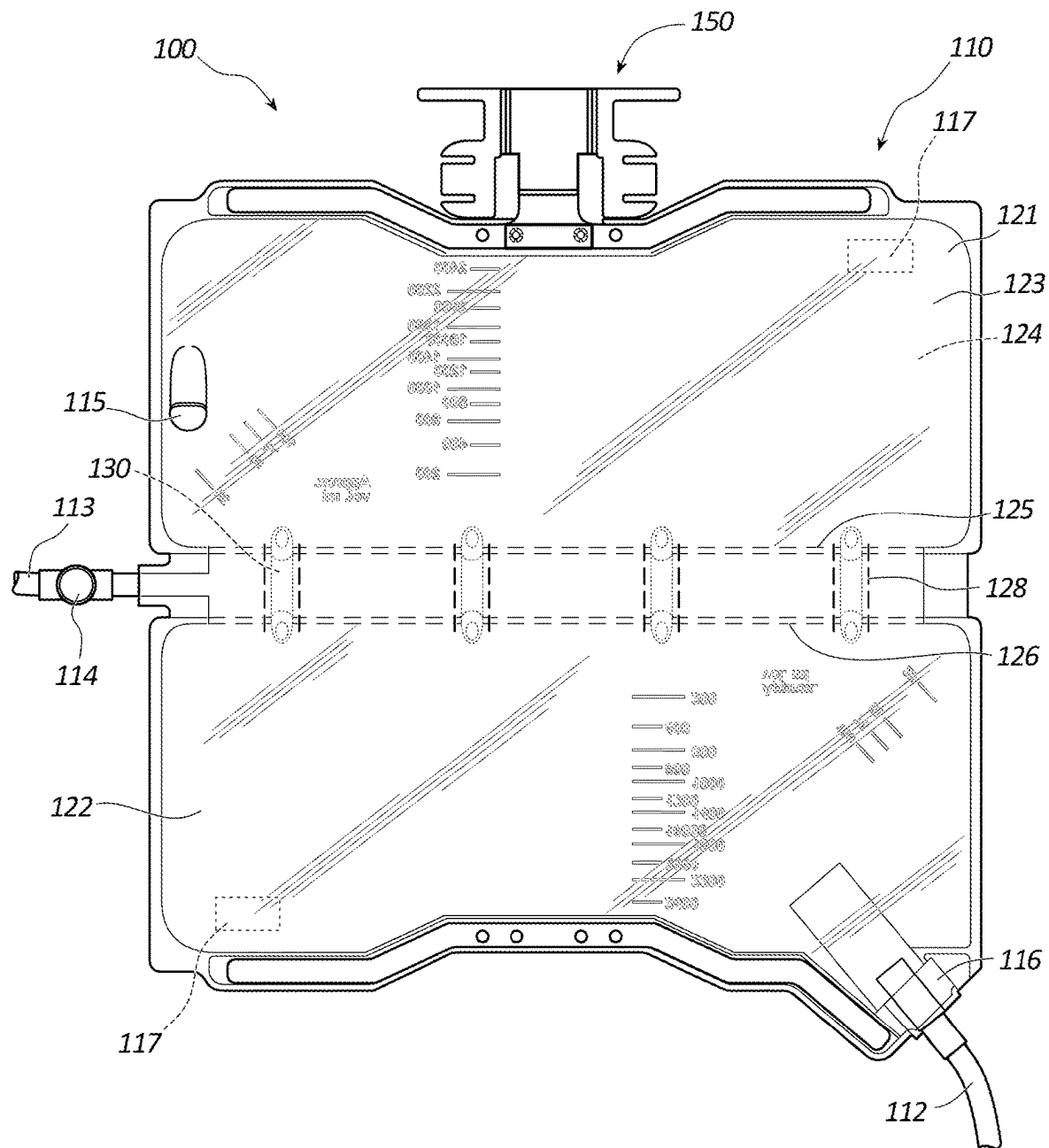
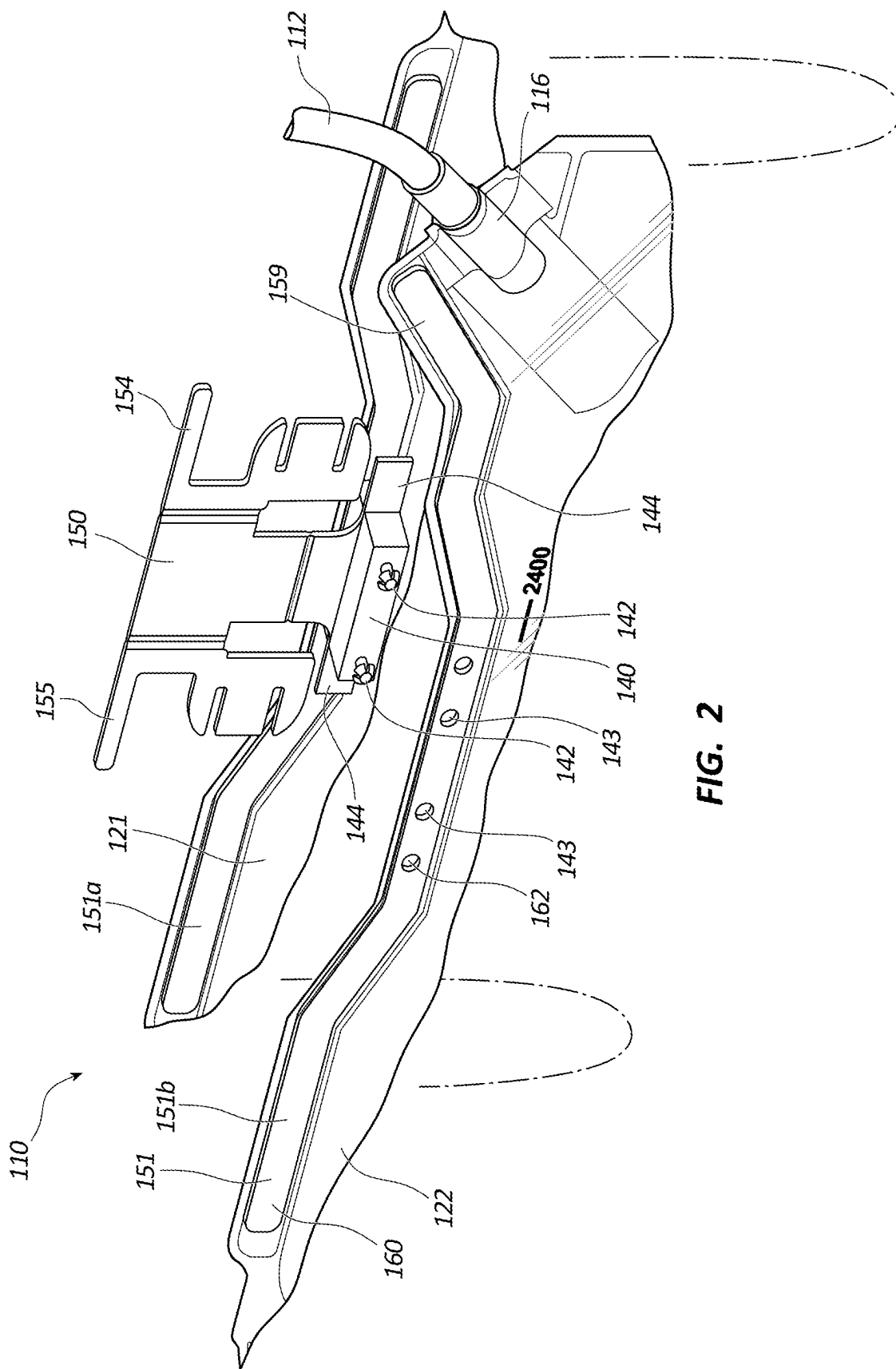
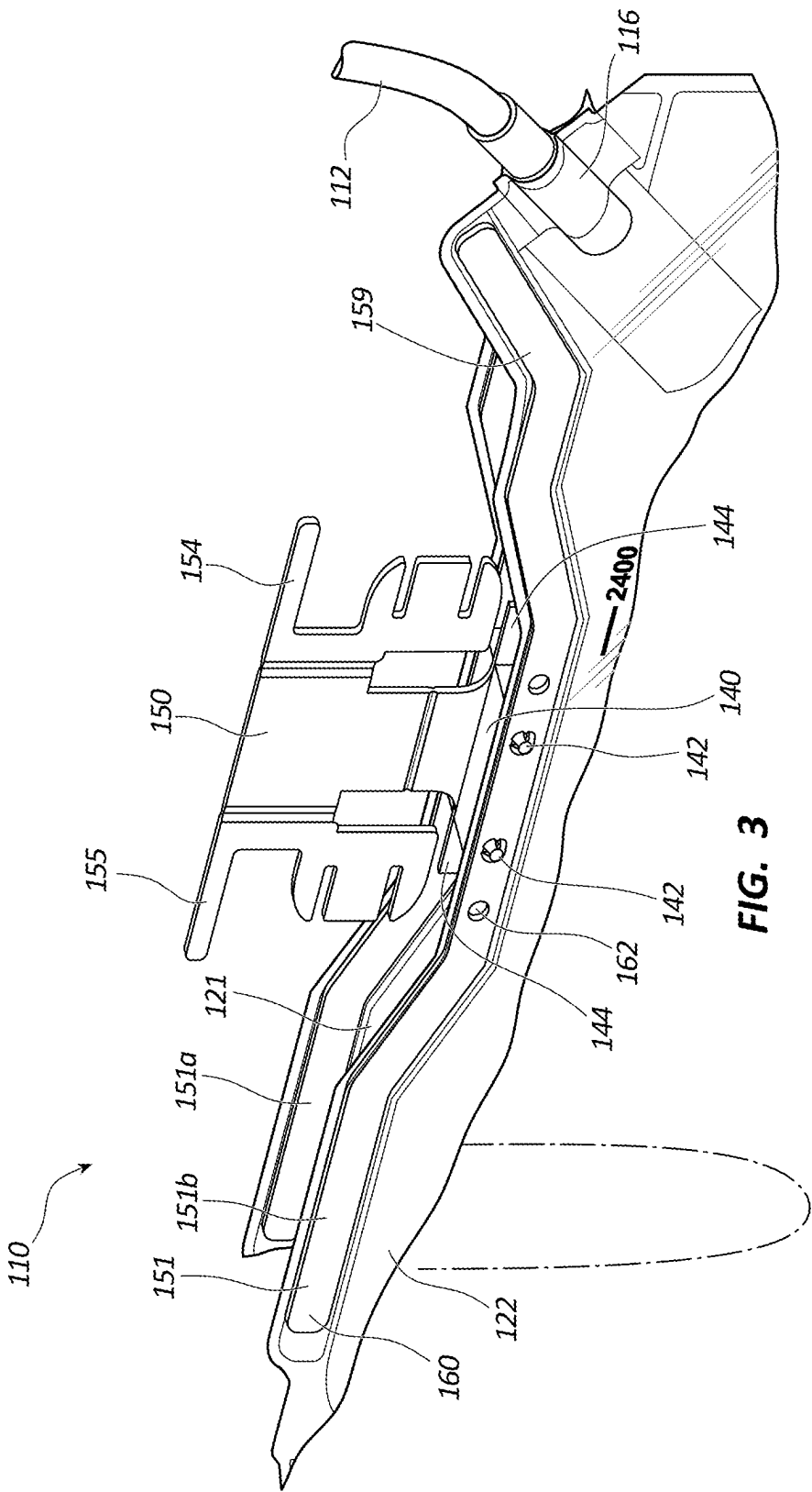
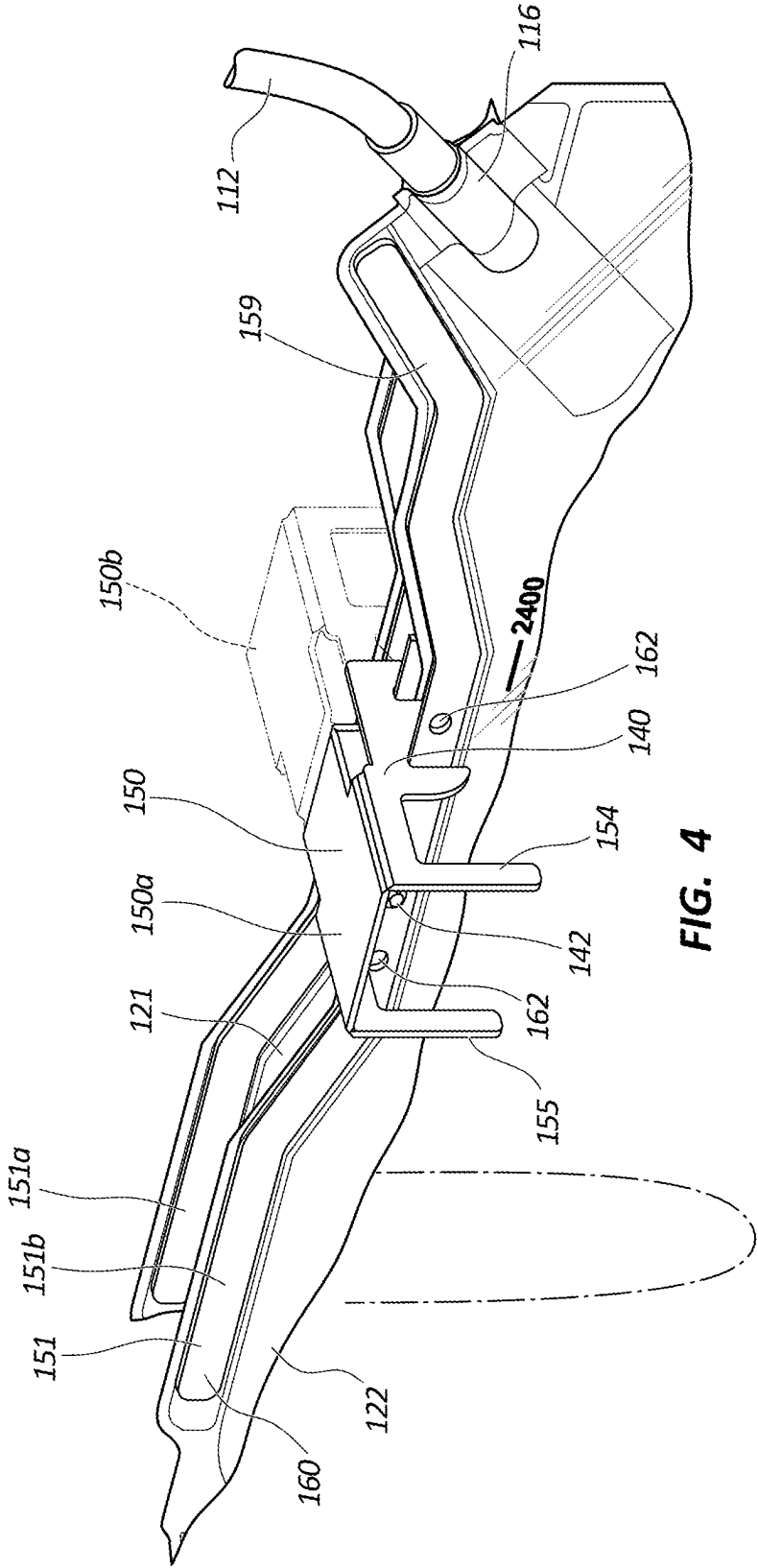
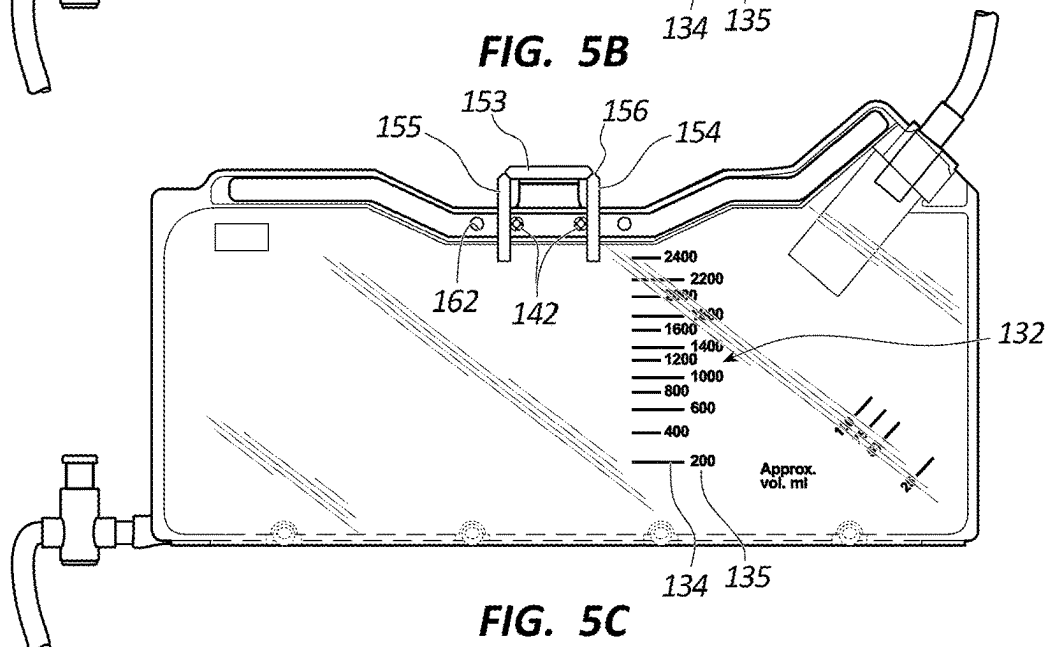
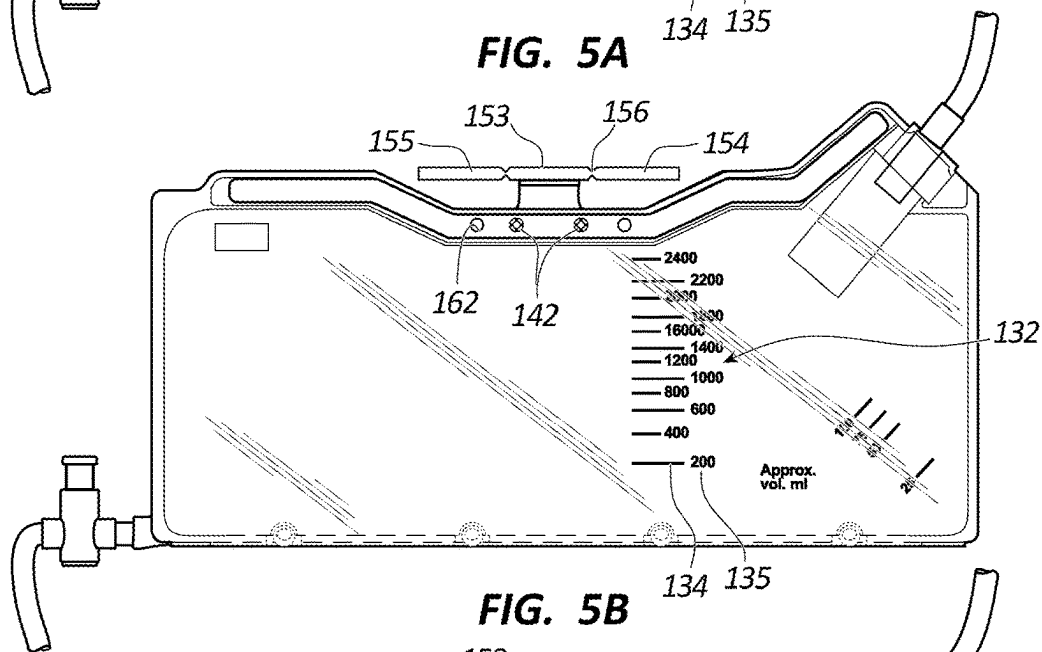
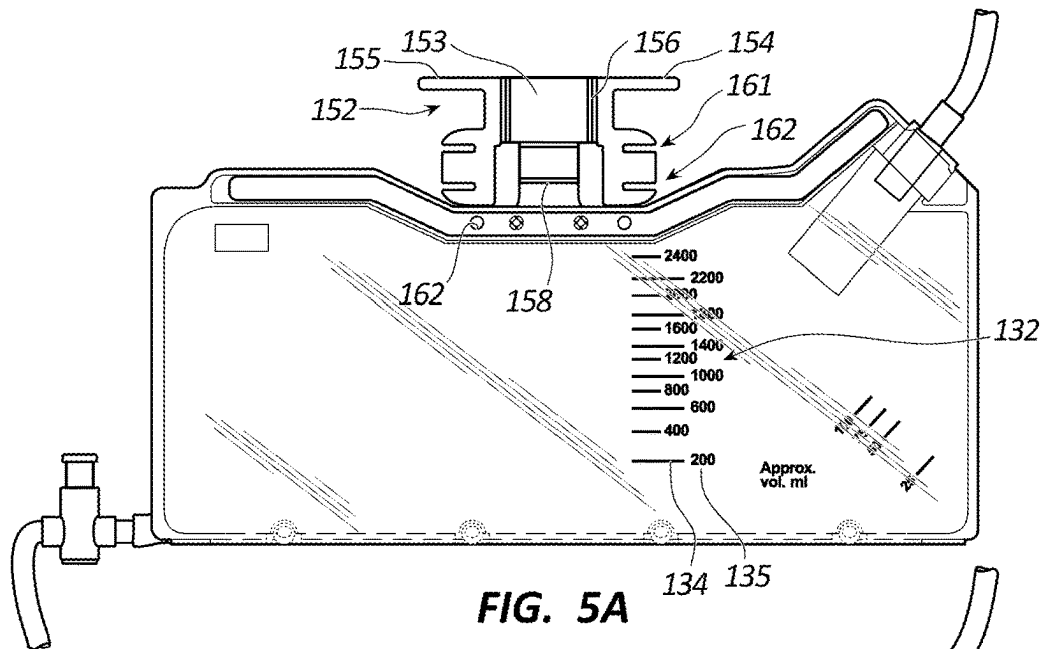


FIG. 1









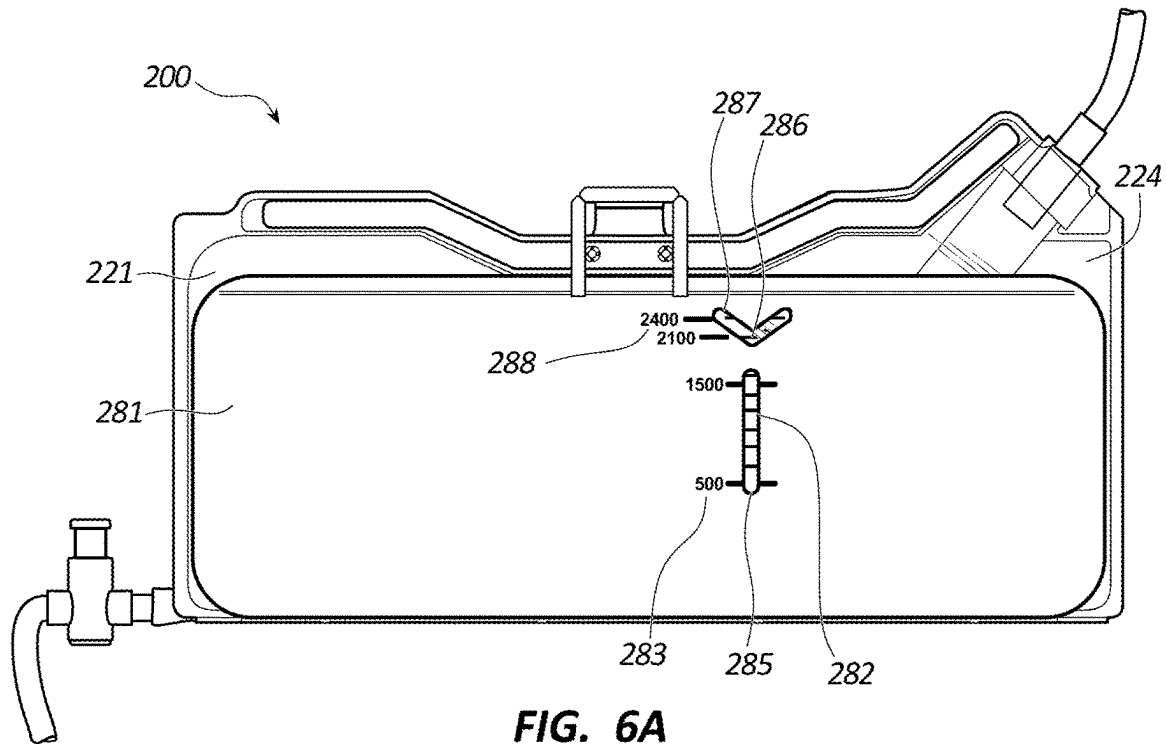


FIG. 6A

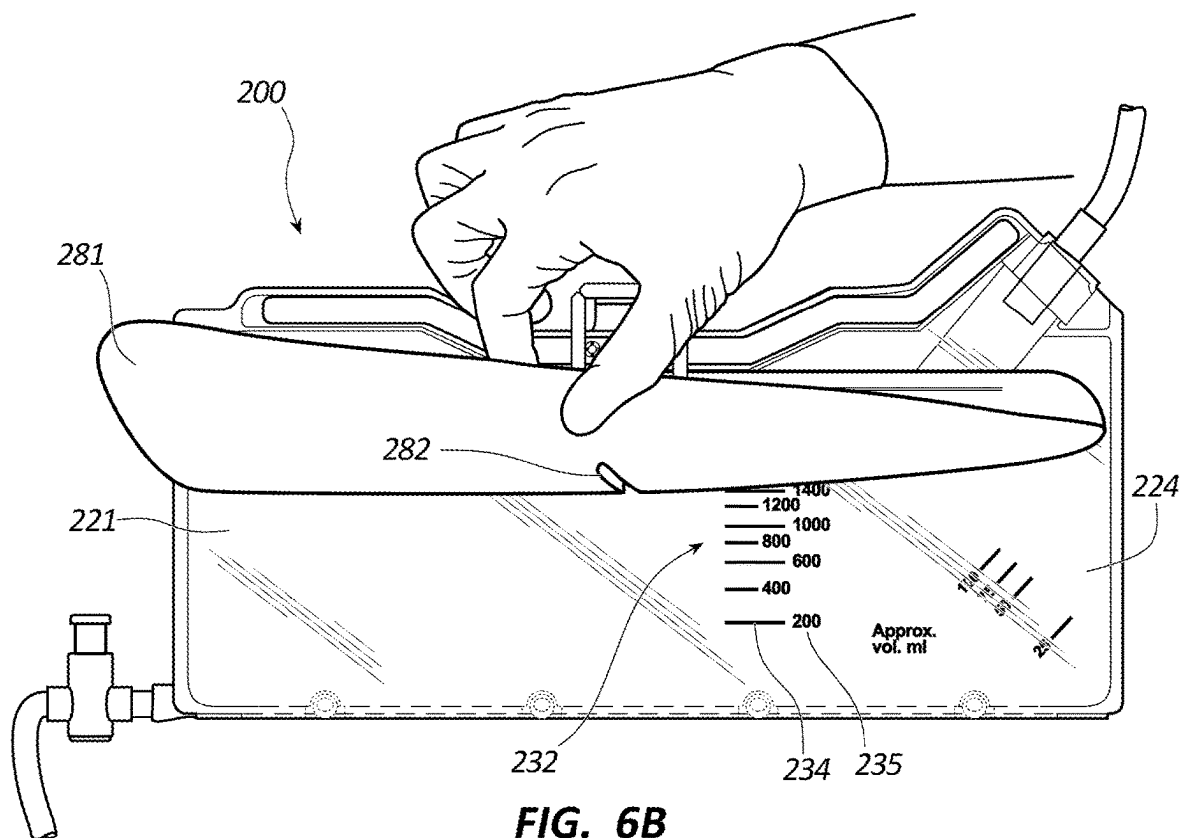


FIG. 6B

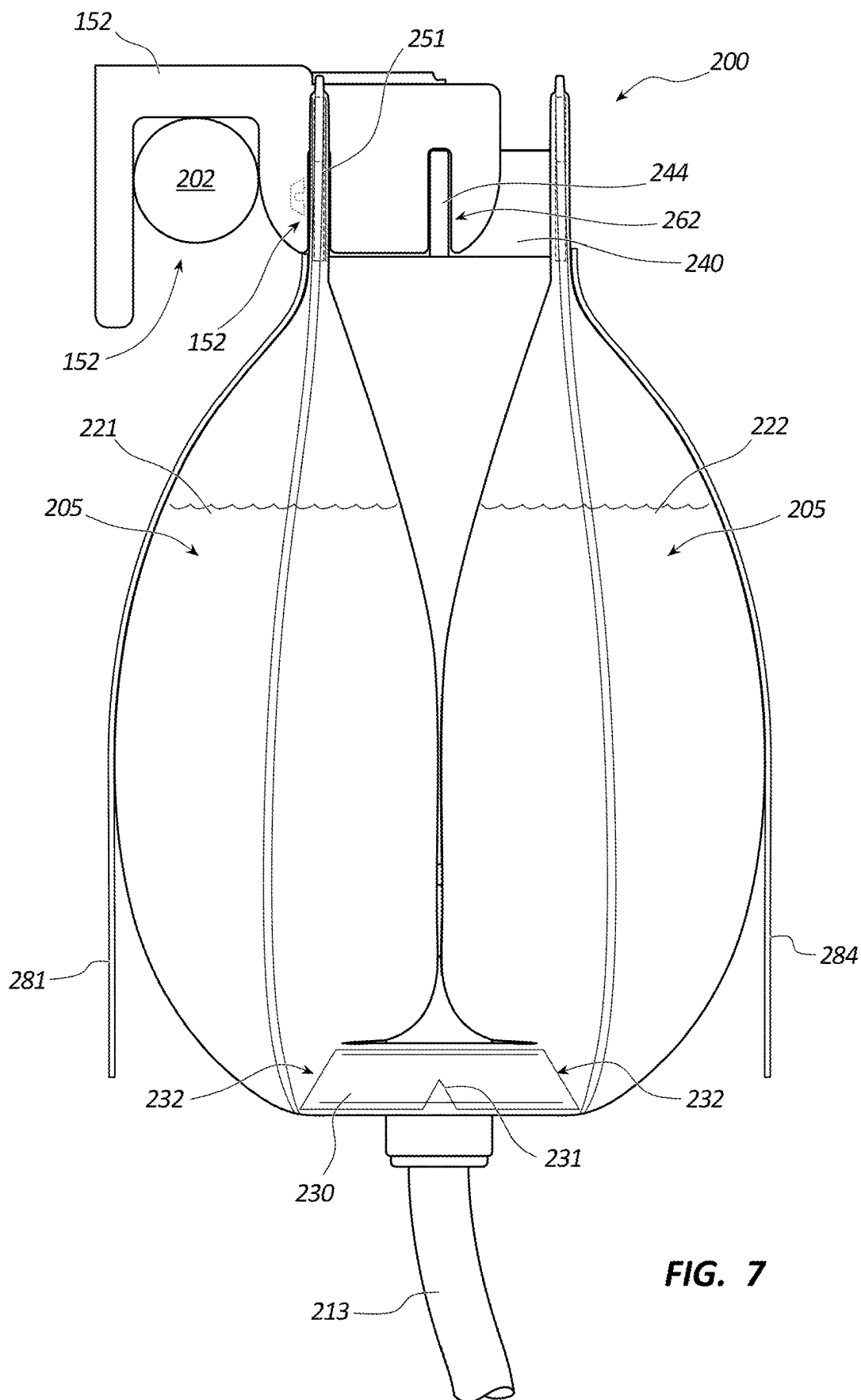


FIG. 7

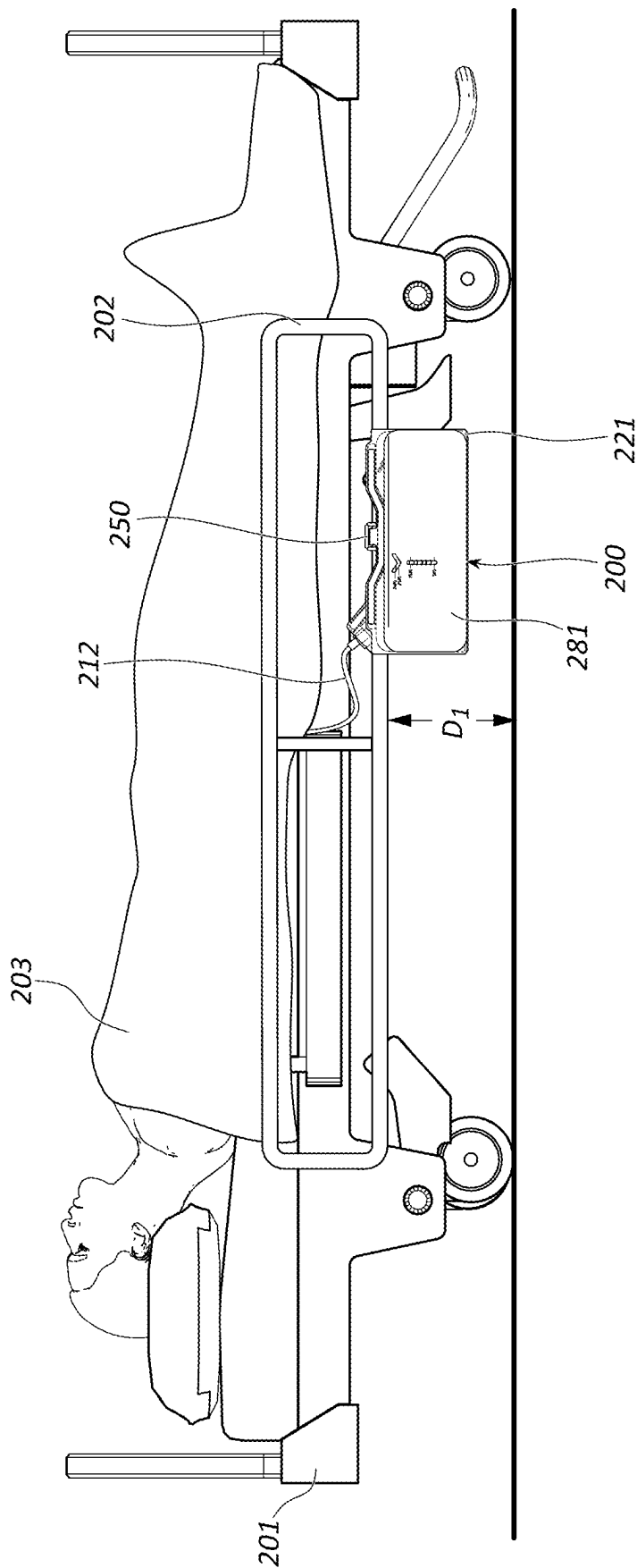


FIG. 8

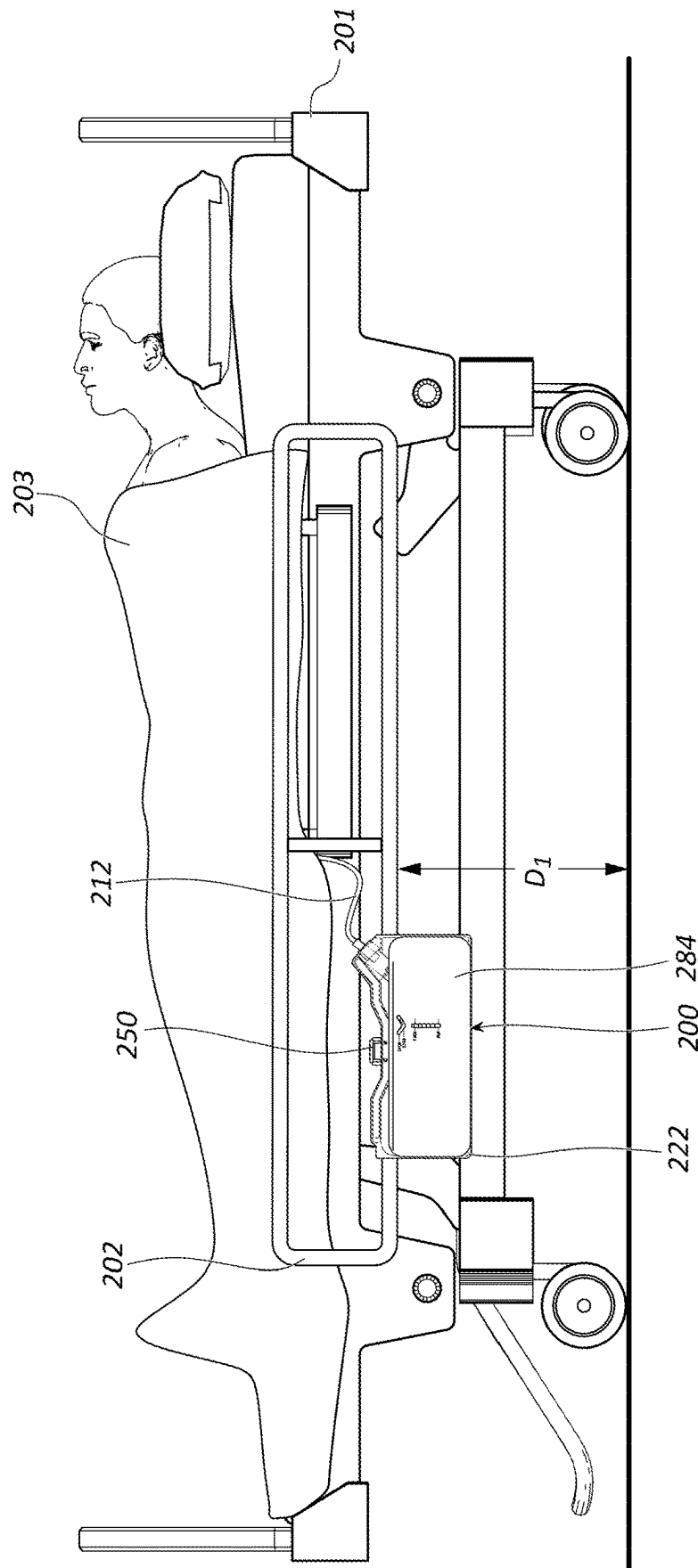


FIG. 9

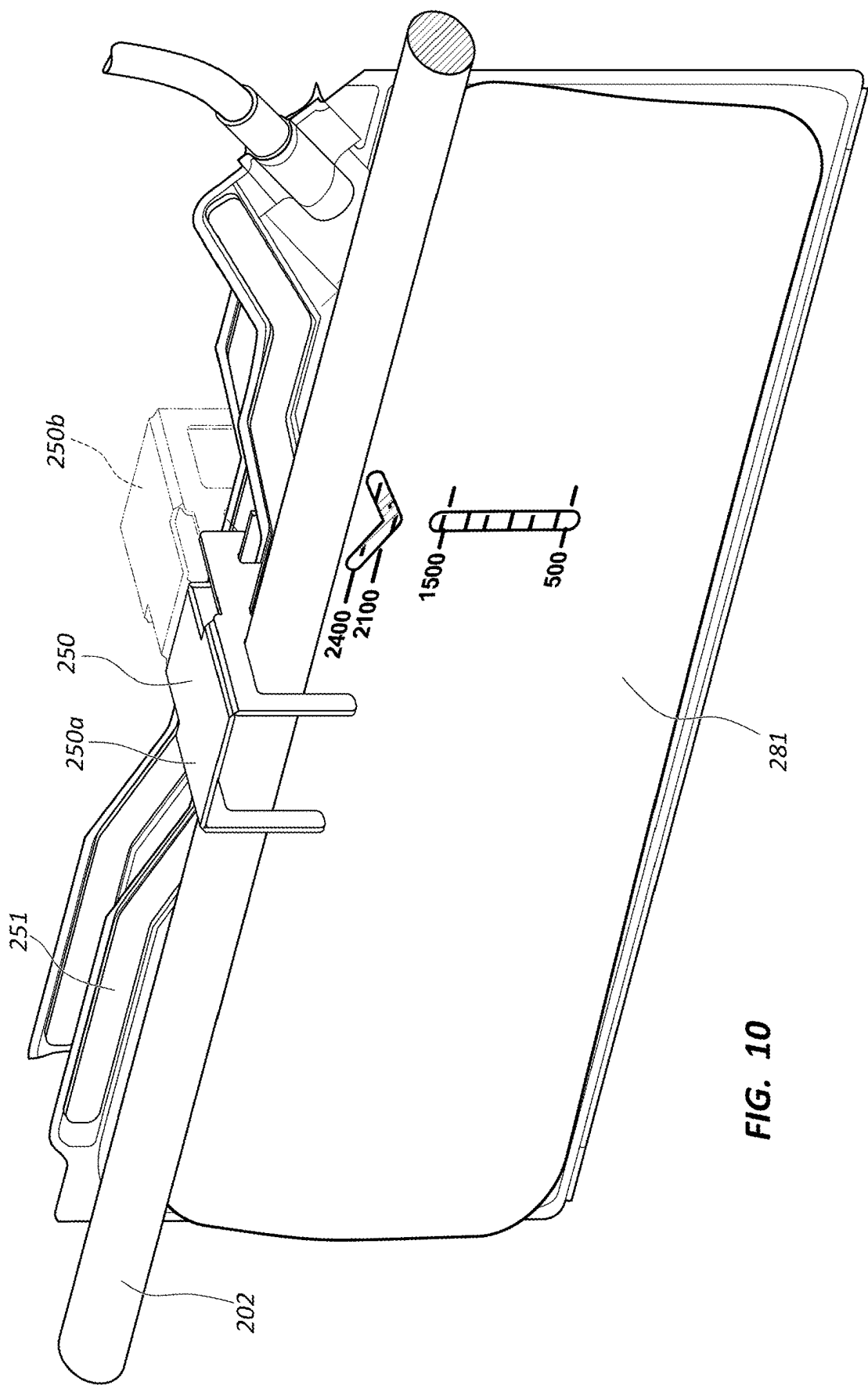


FIG. 10

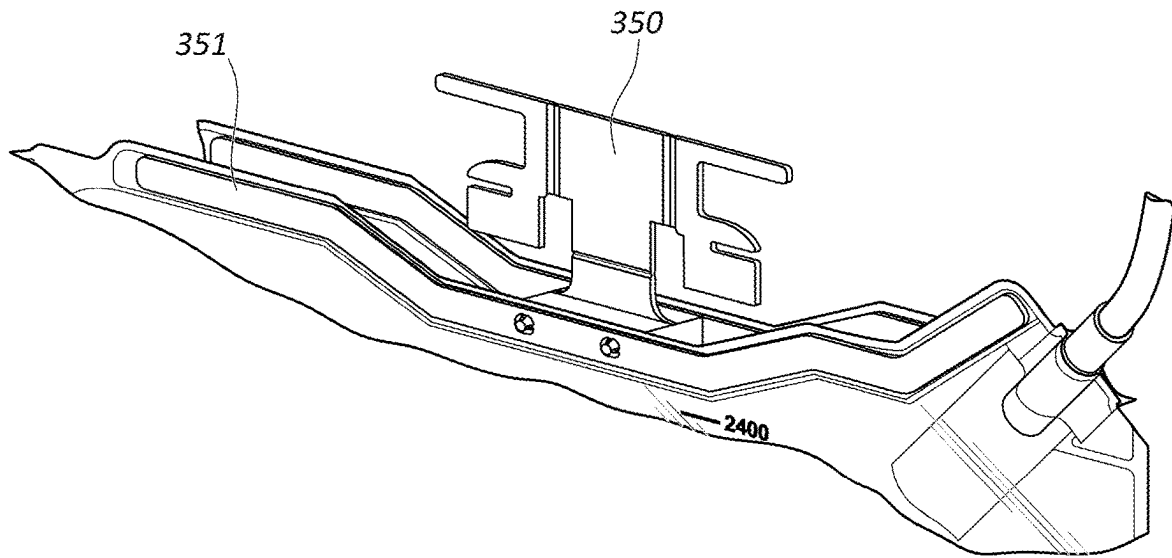


FIG. 11A

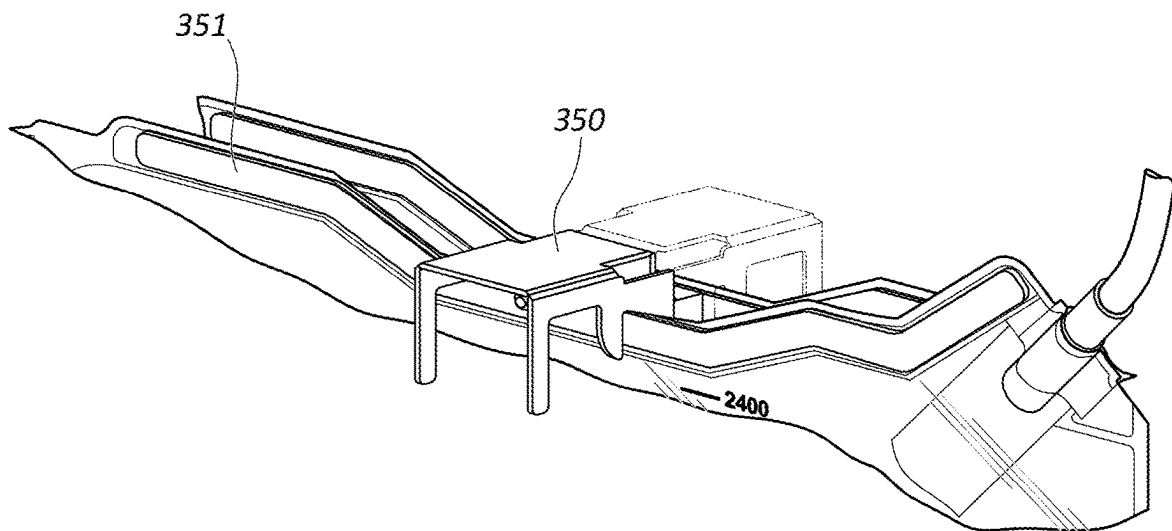


FIG. 11B

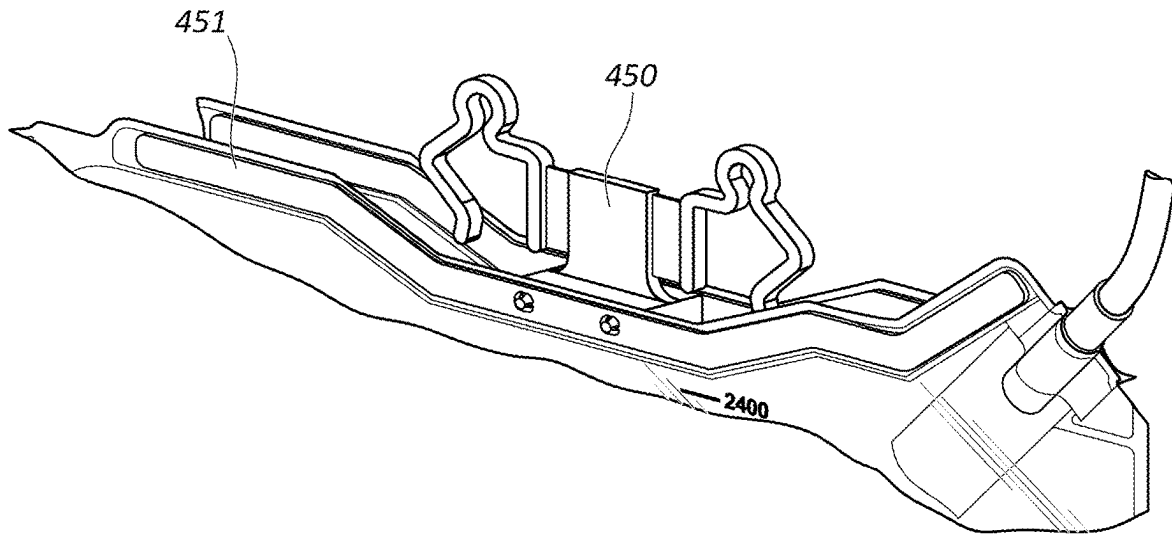


FIG. 12A

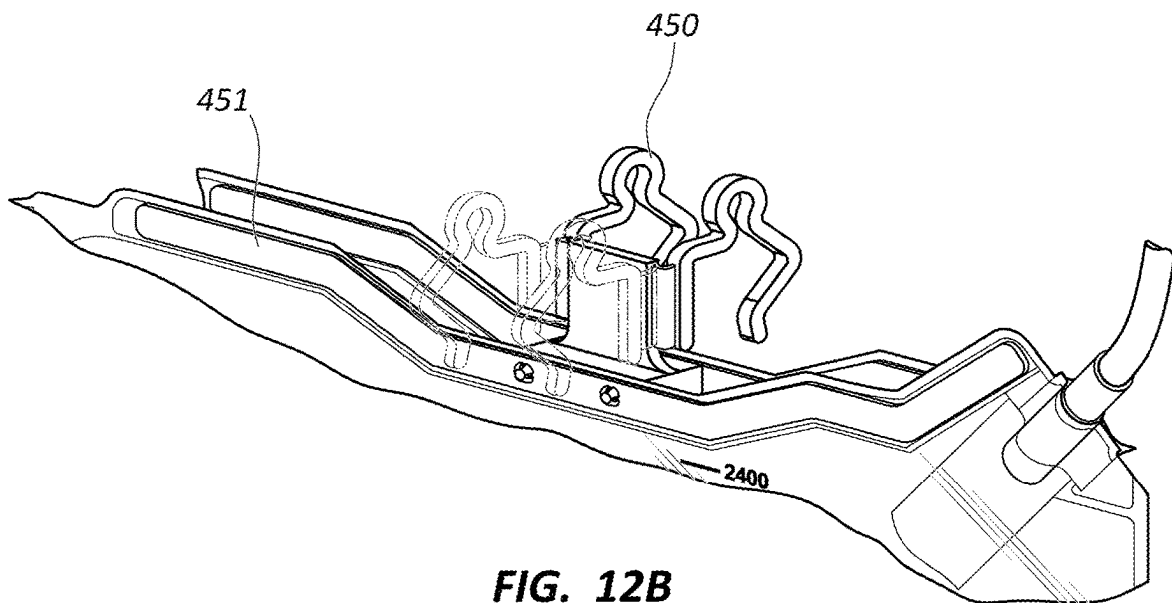
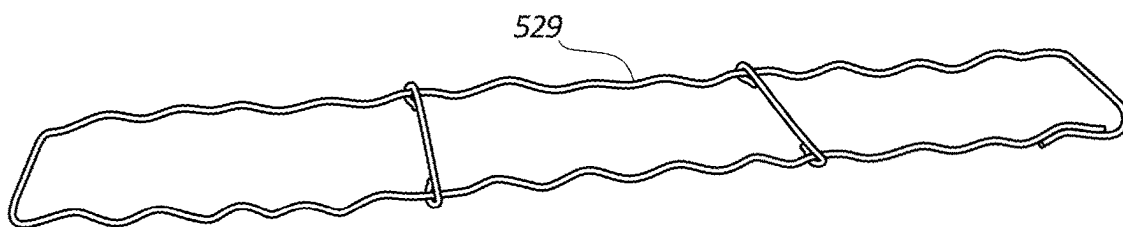
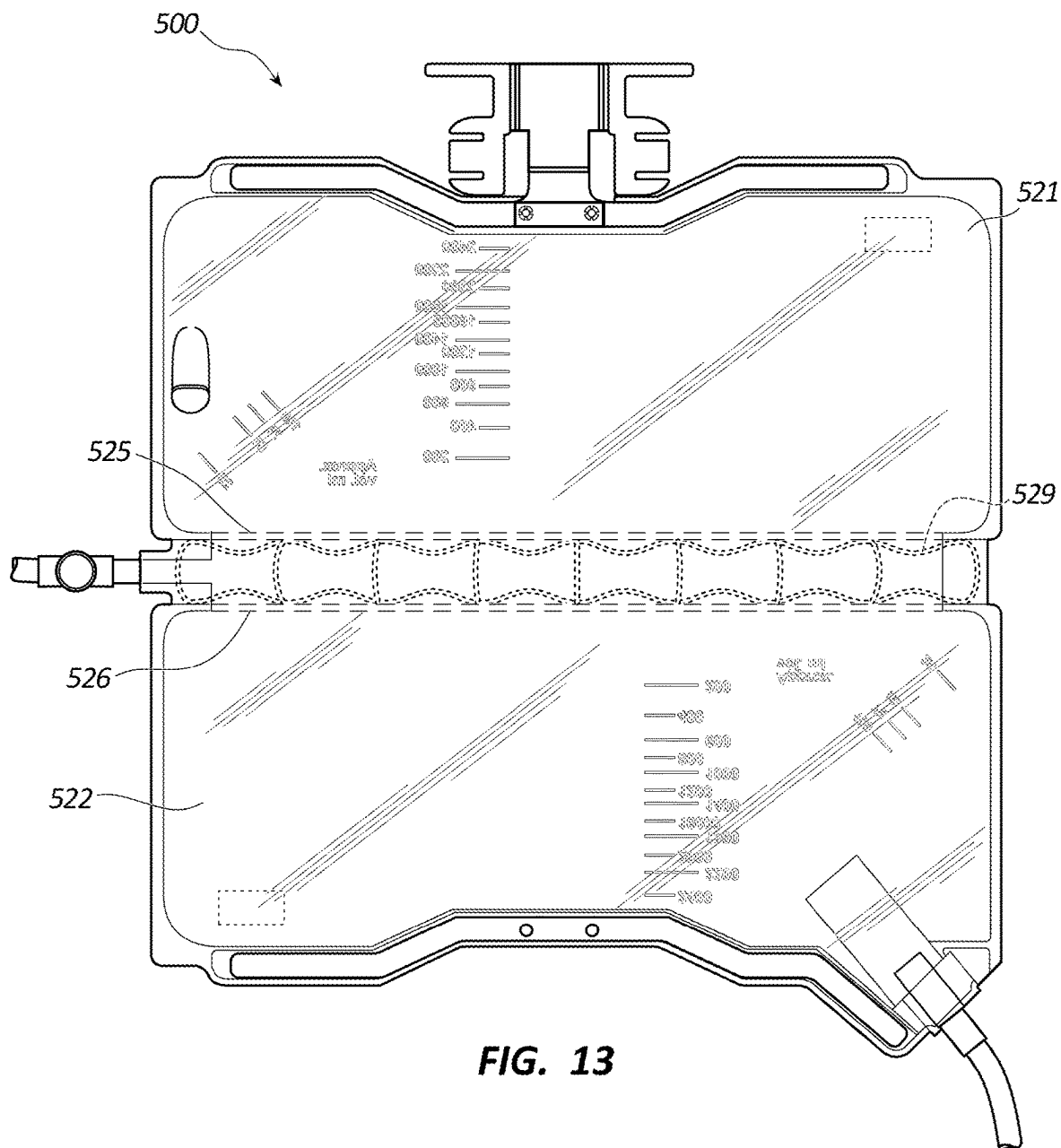


FIG. 12B



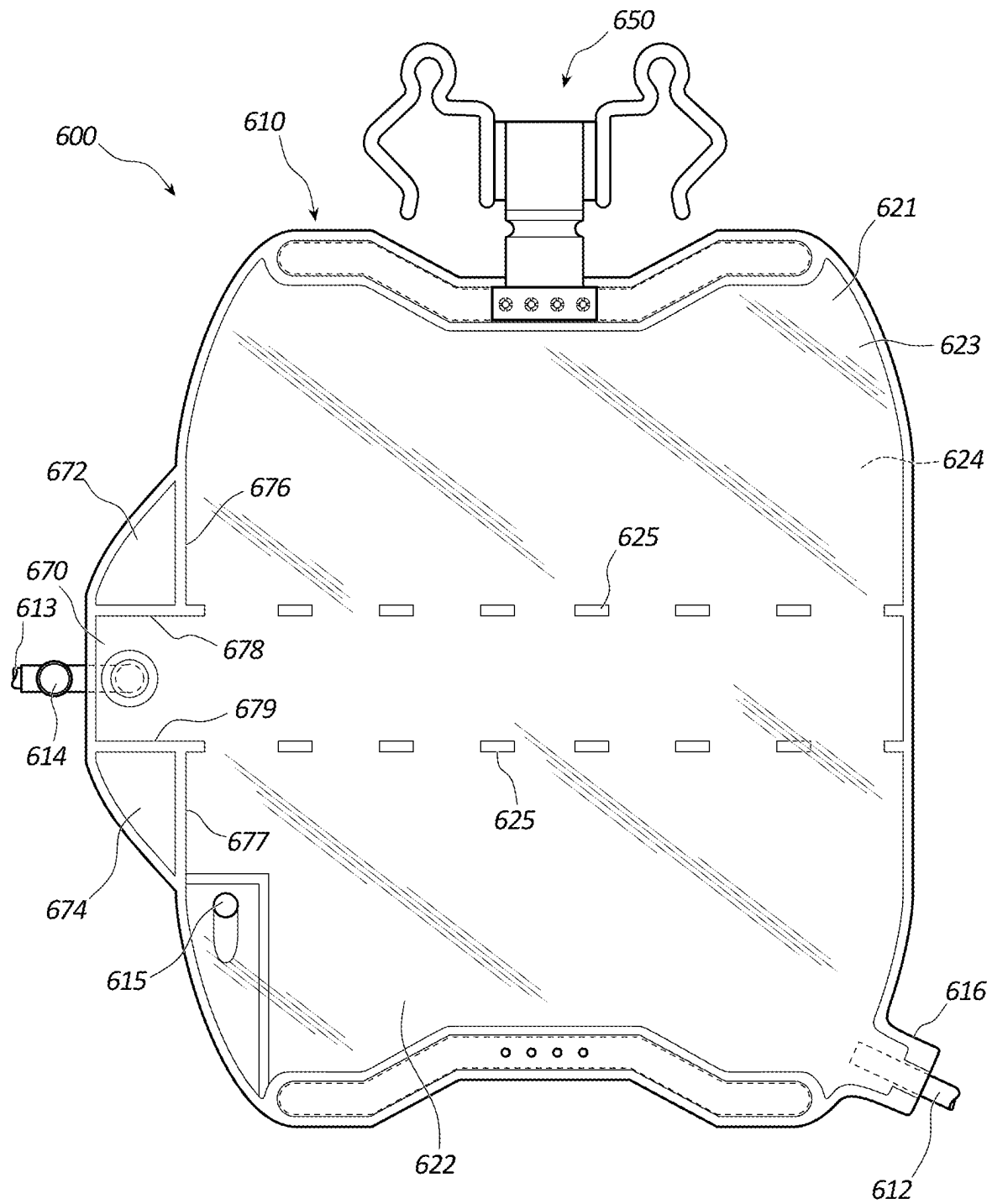


FIG. 15

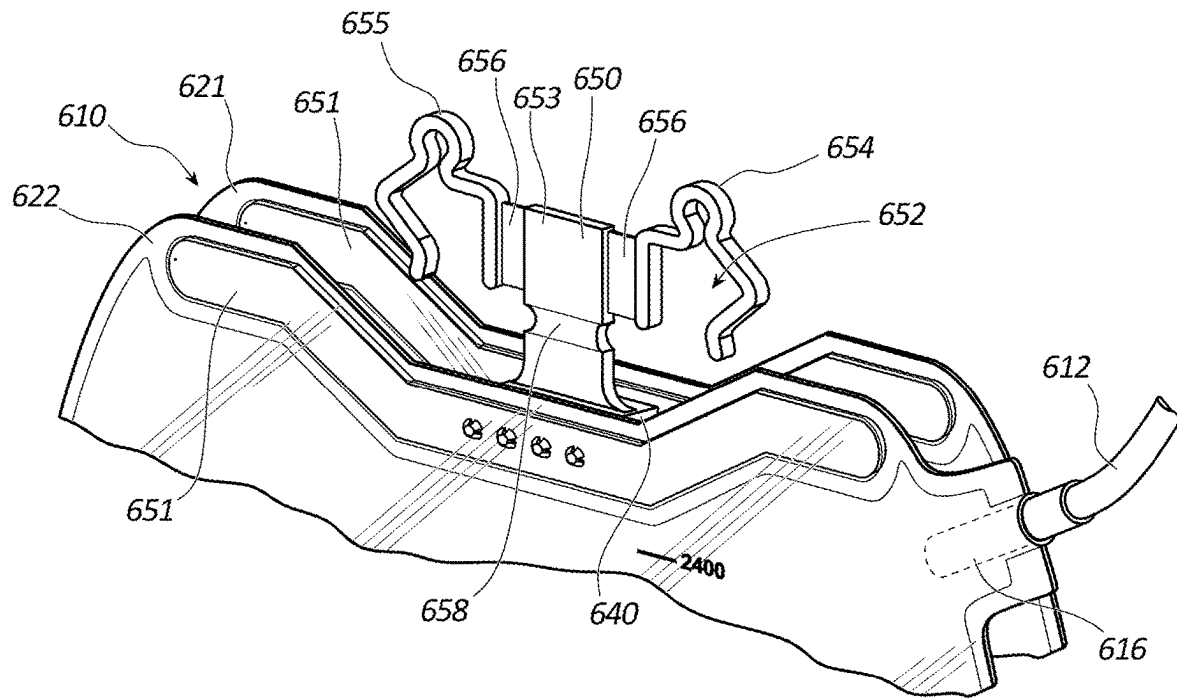


FIG. 16A

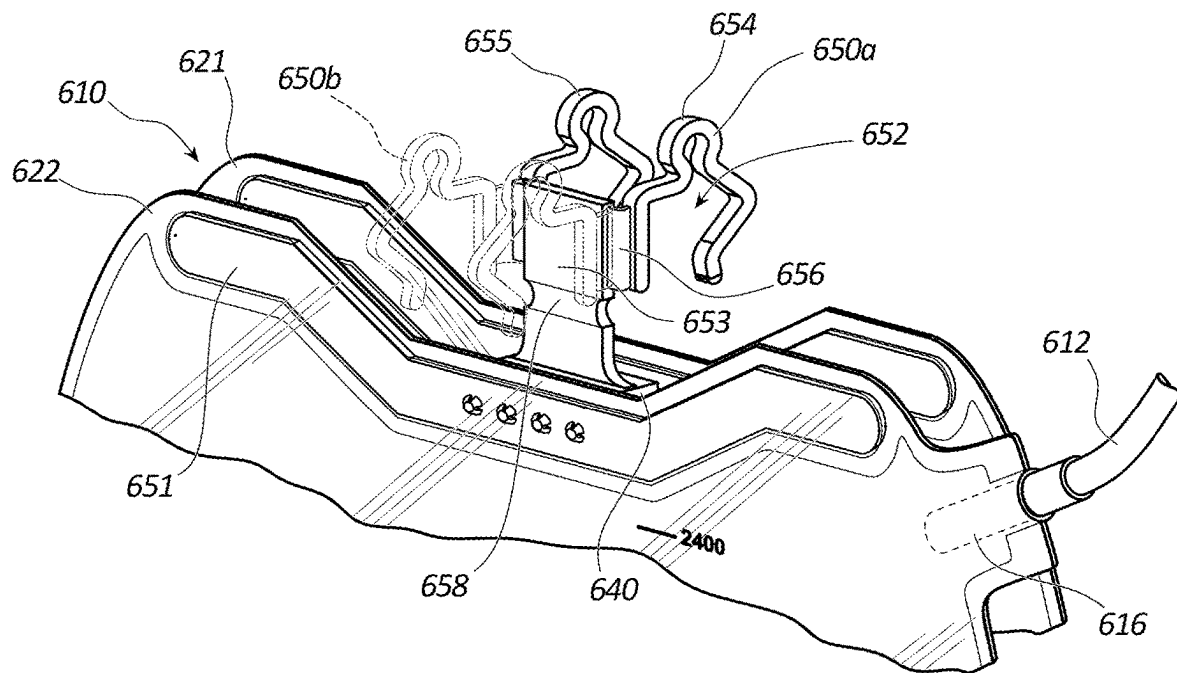


FIG. 16B

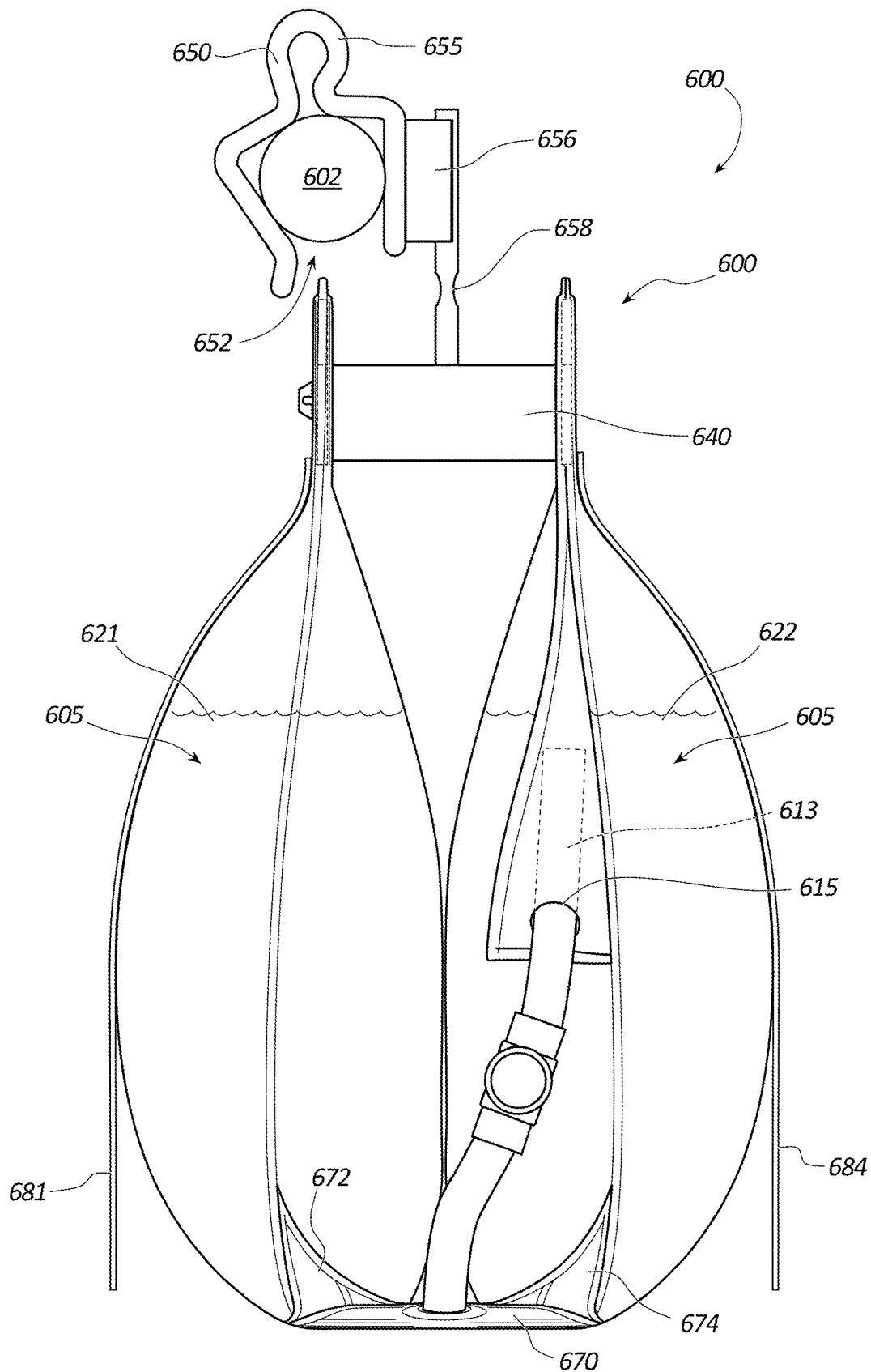
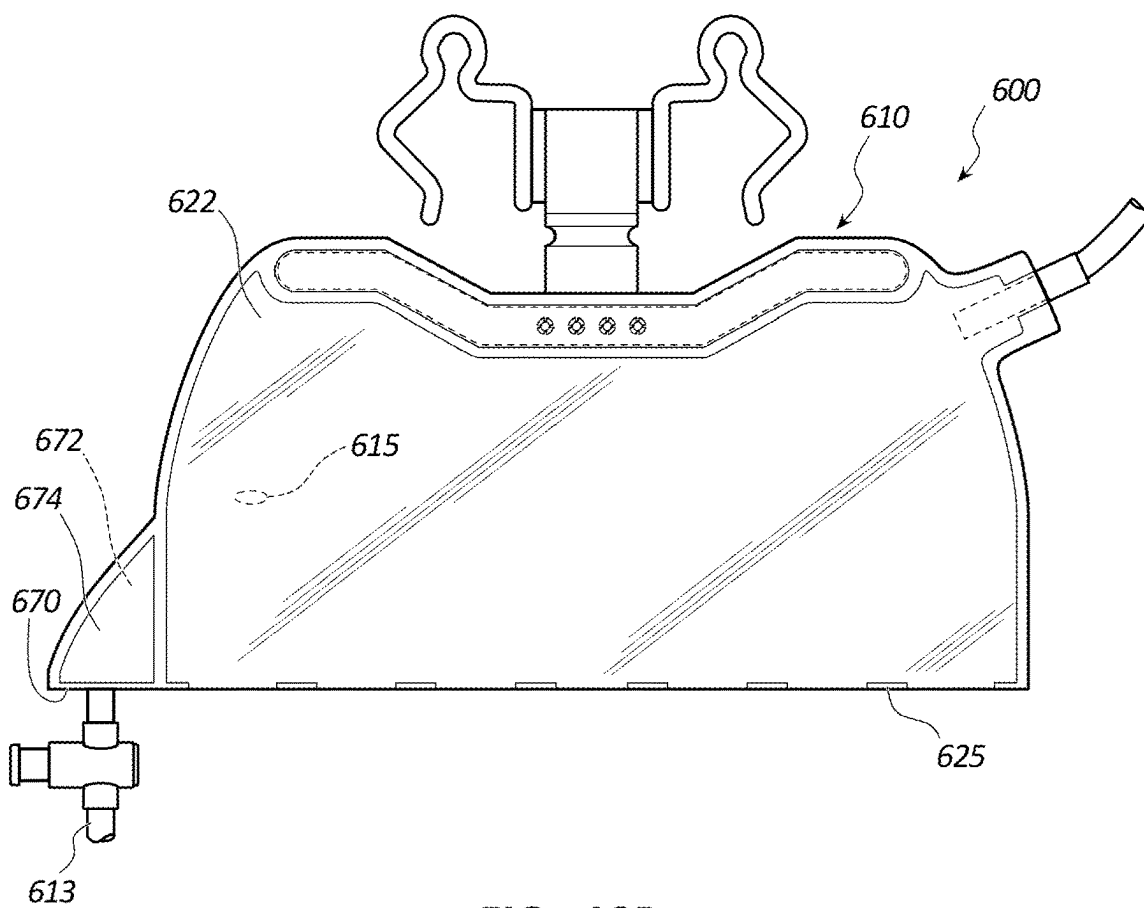
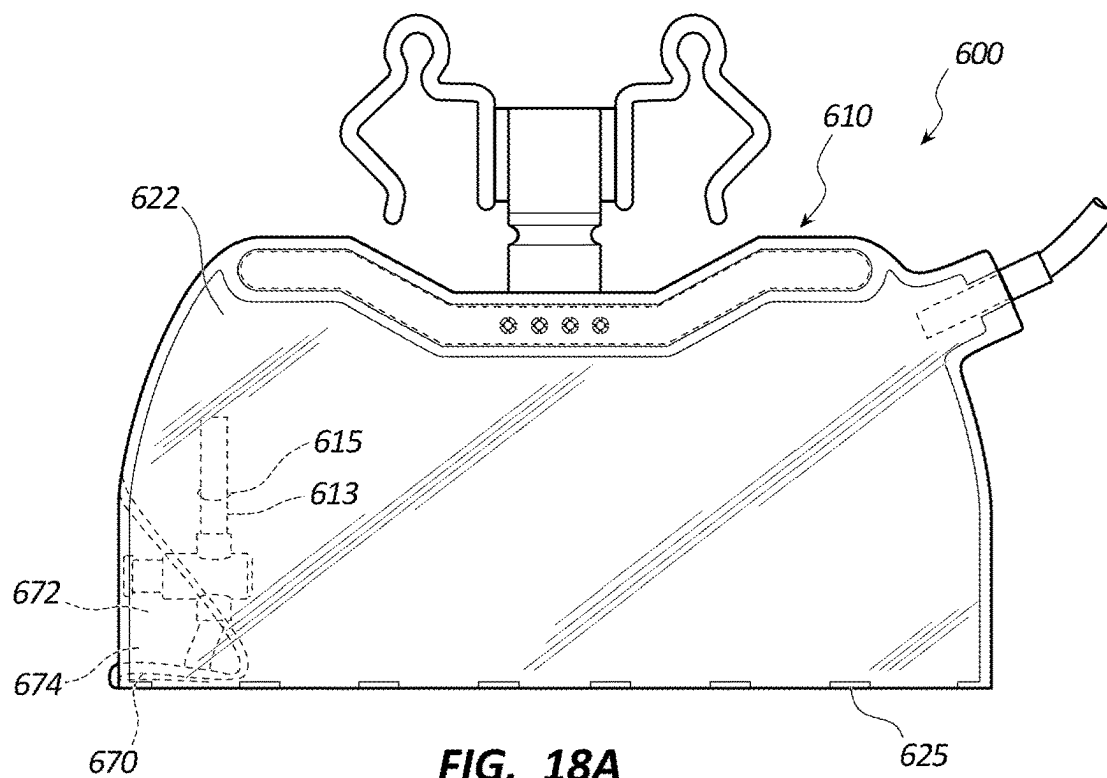
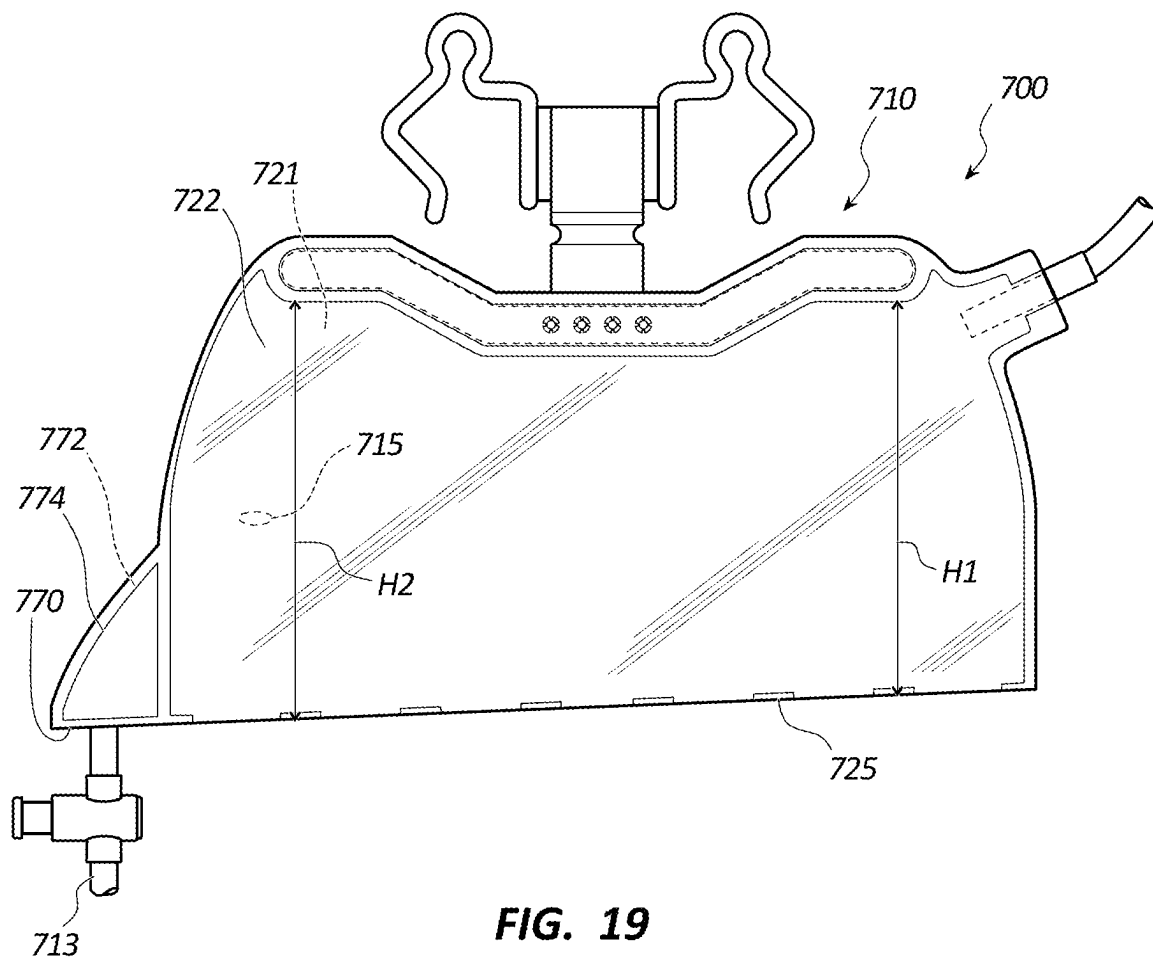


FIG. 17





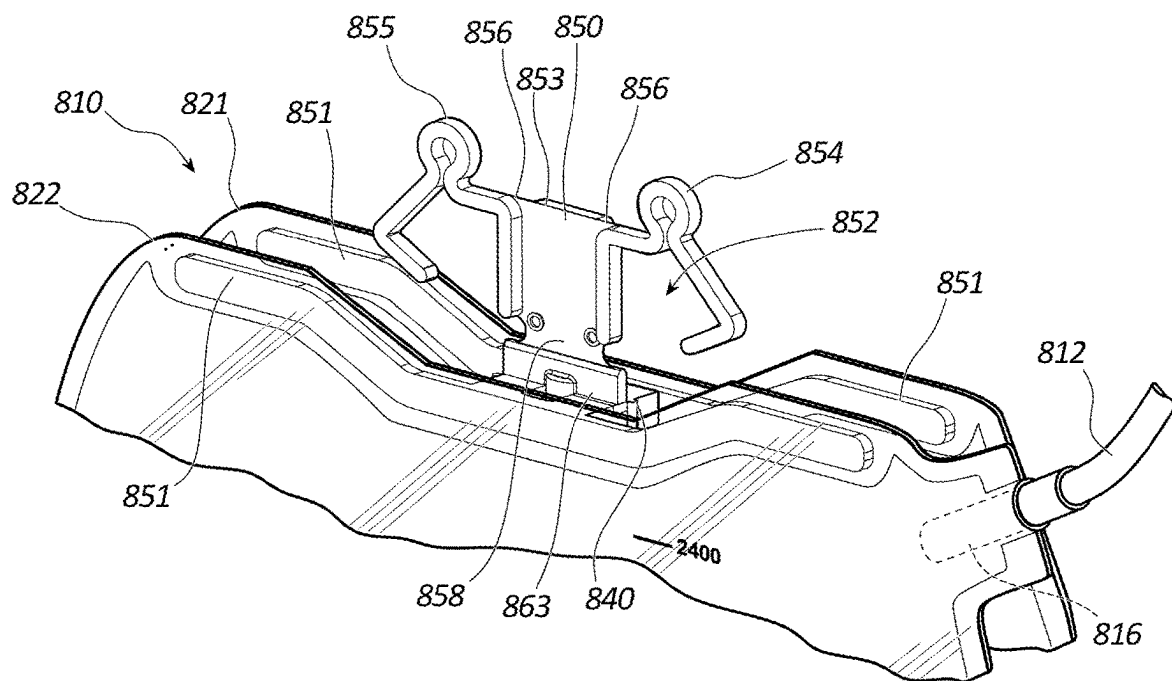


FIG. 20A

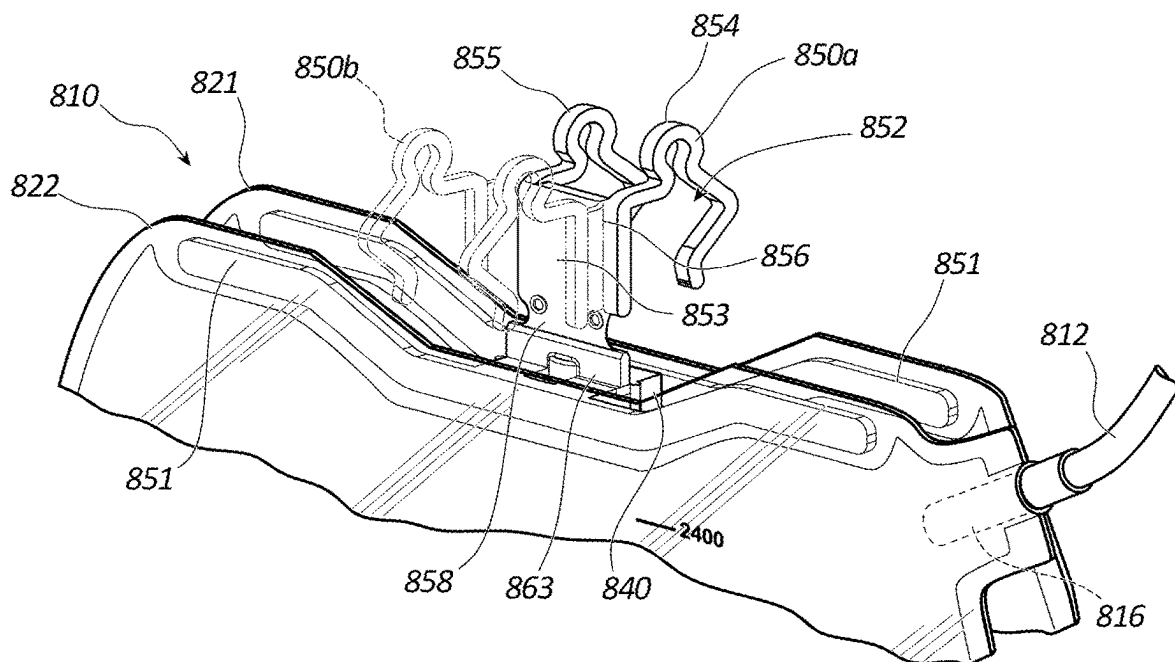


FIG. 20B

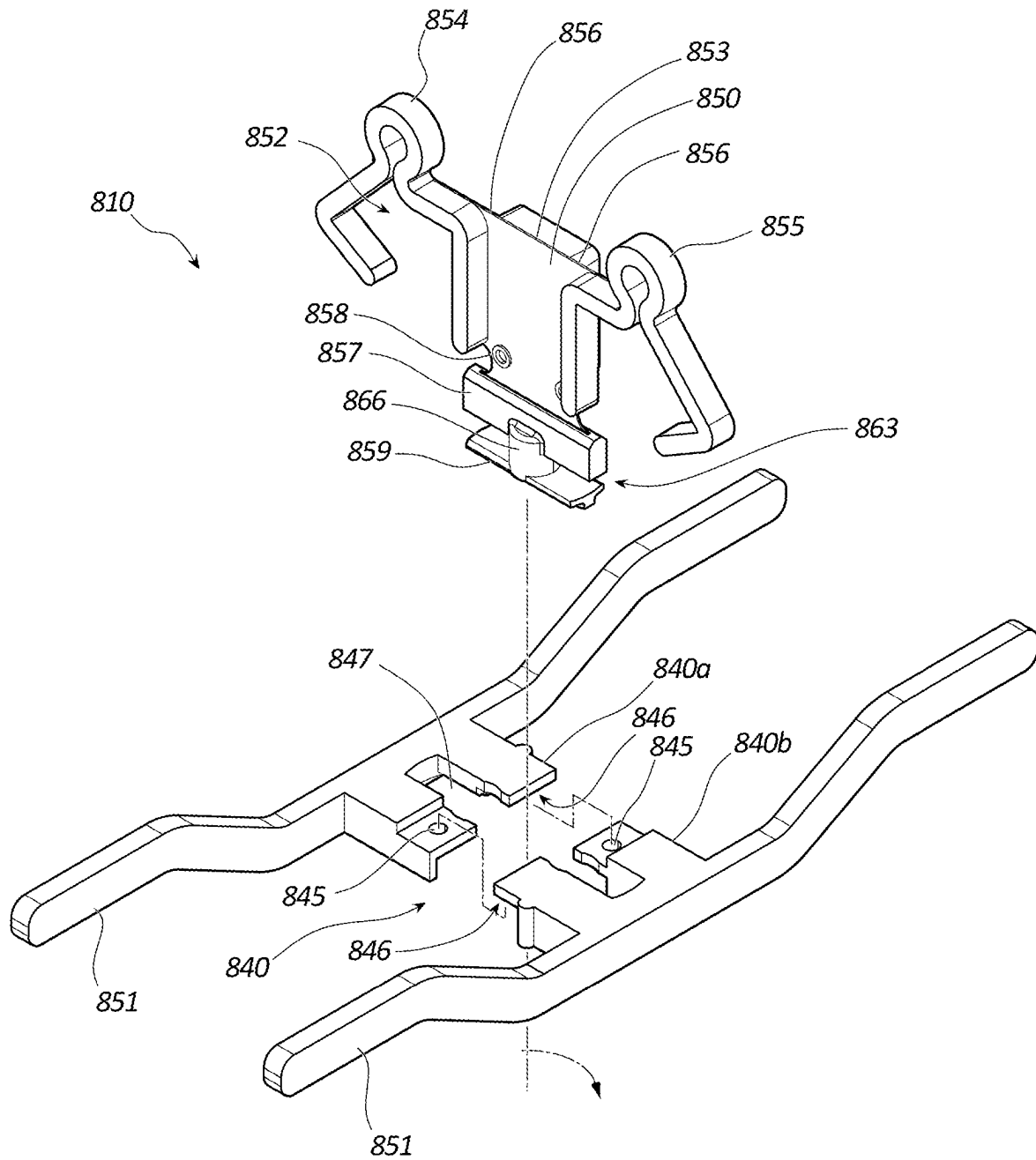


FIG. 21

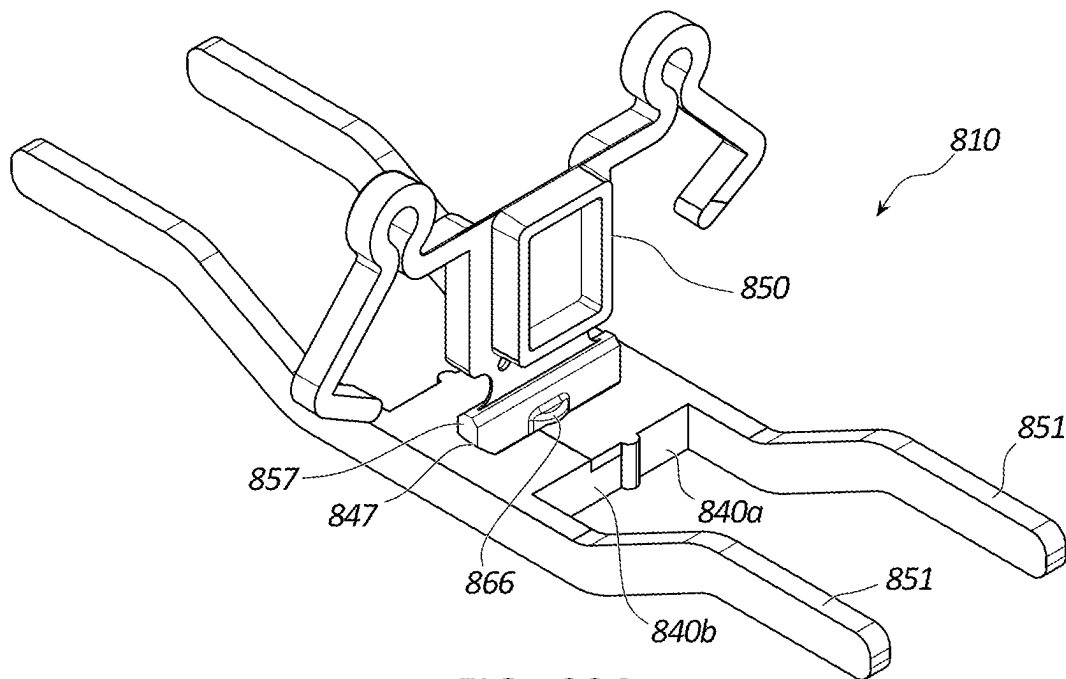


FIG. 22A

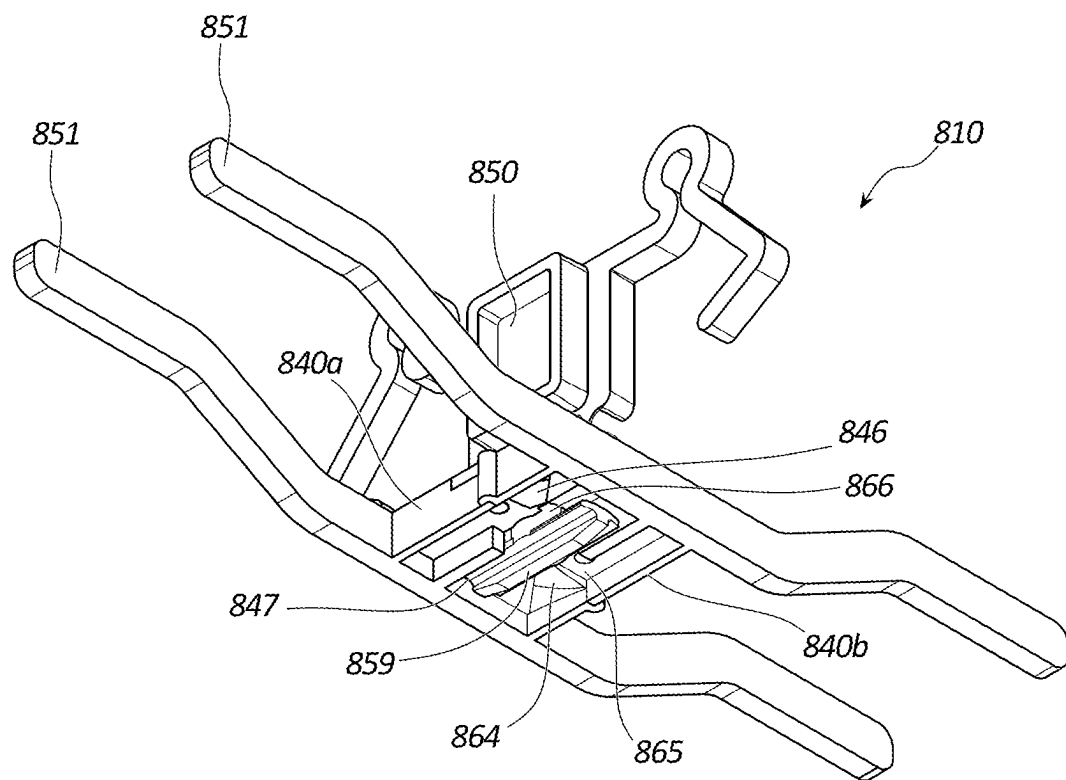


FIG. 22B

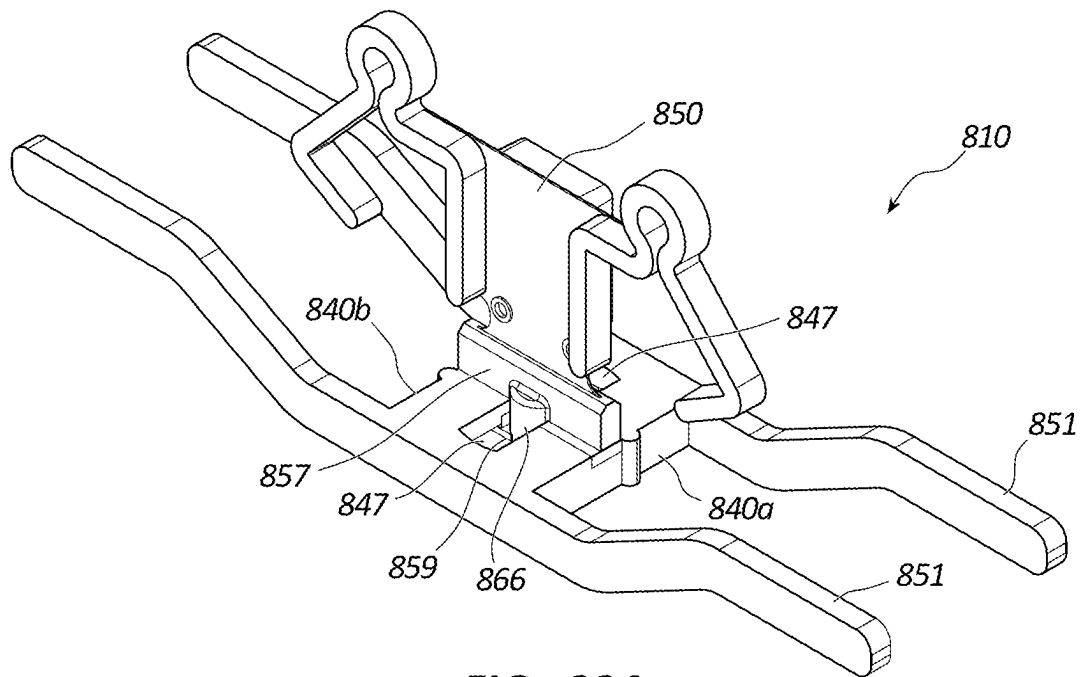


FIG. 23A

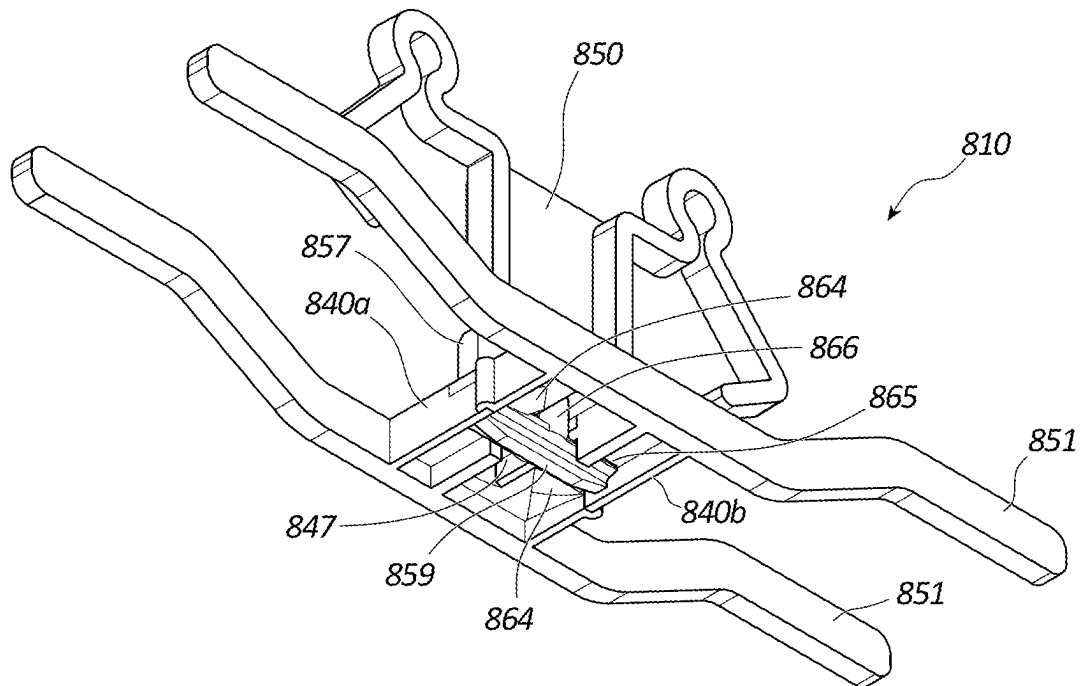


FIG. 23B

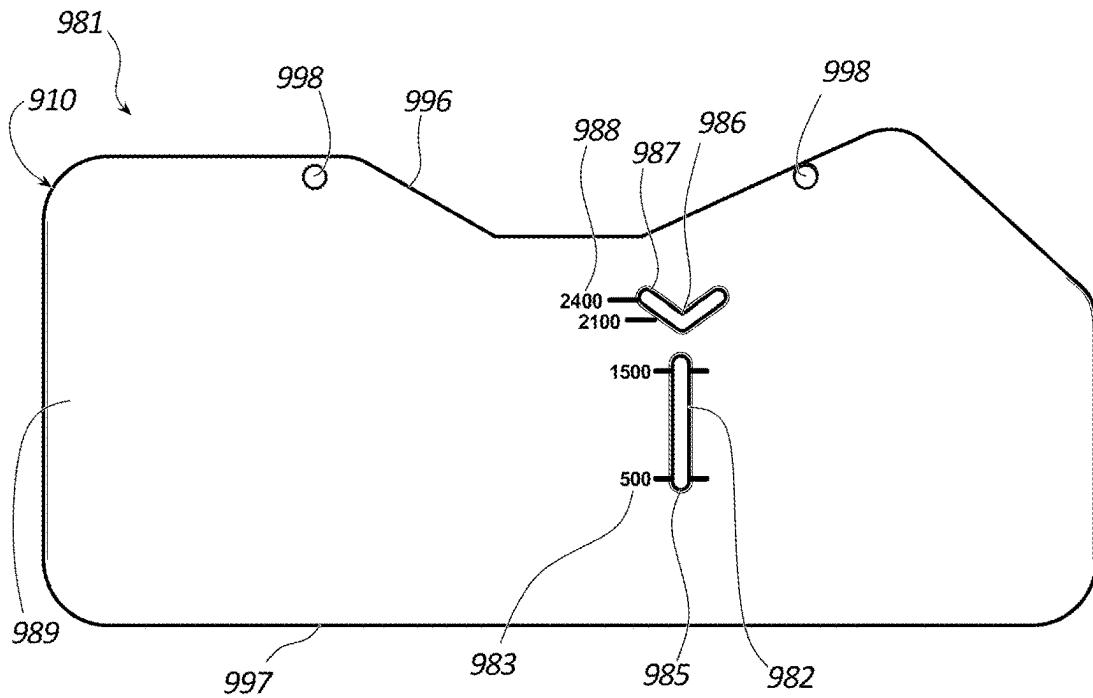


FIG. 24A

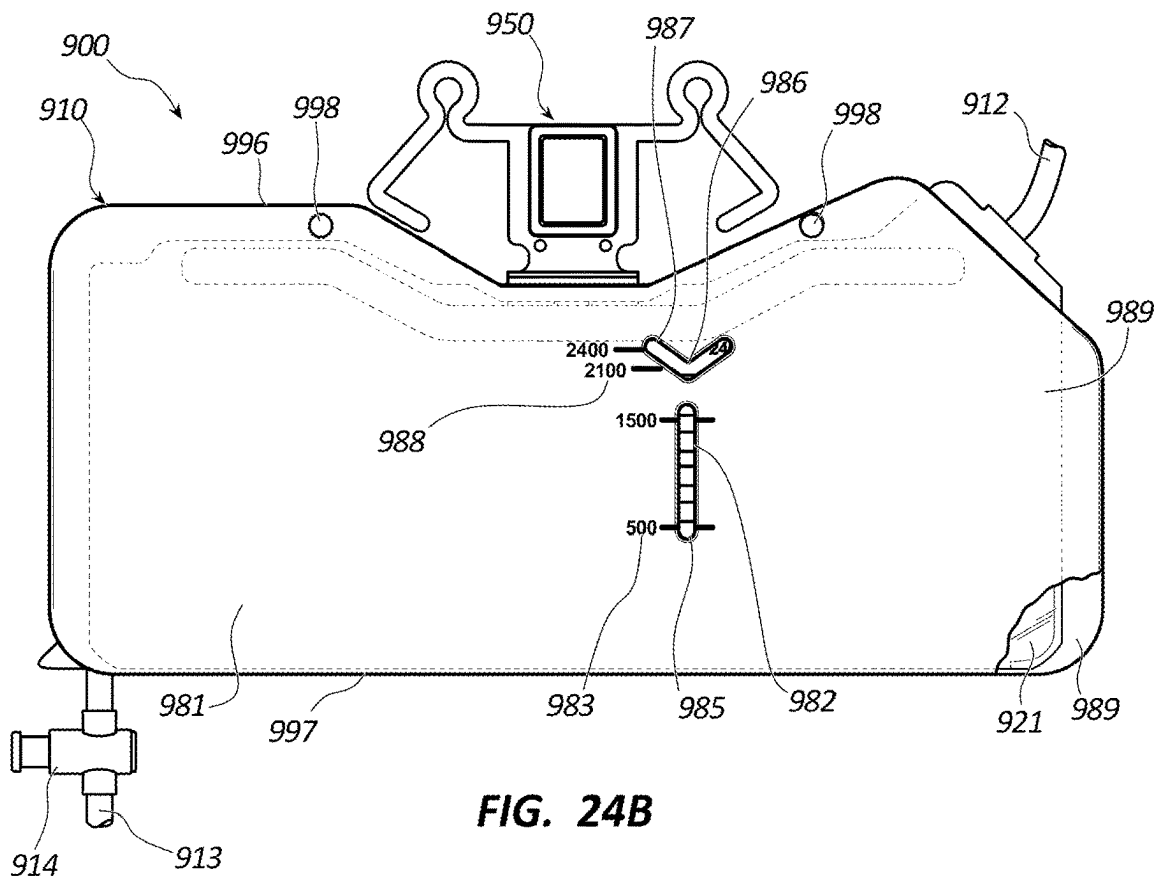


FIG. 24B

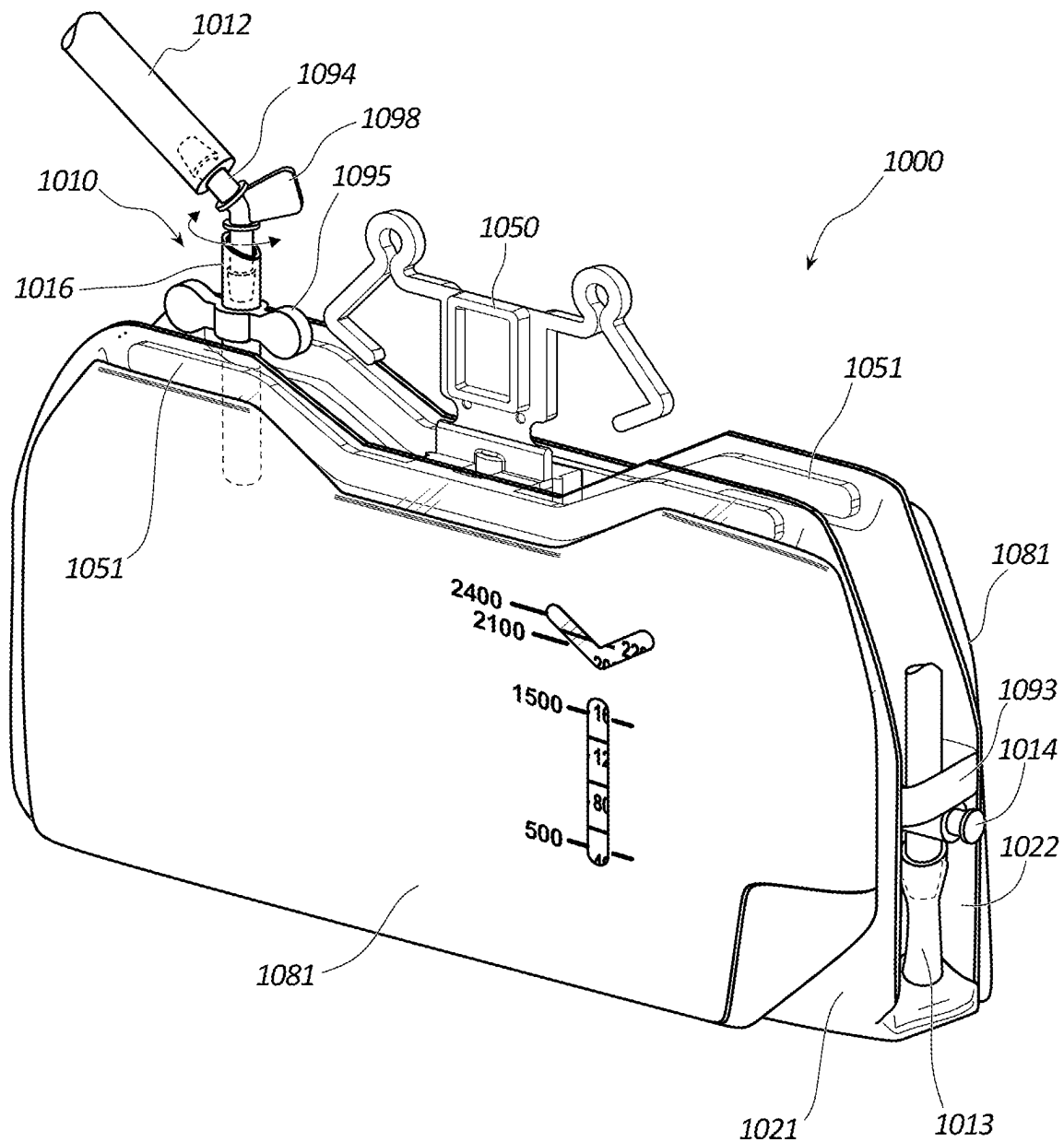


FIG. 25

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DUAL DRAINAGE BAG, ASSEMBLIES, AND RELATED METHODS

RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 16/836,585, filed Mar. 31, 2020, and titled DUAL DRAINAGE BAG, ASSEMBLIES, AND RELATED METHODS, which claims priority to U.S. Provisional Patent Application No. 62/827,724, filed Apr. 1, 2019, and titled DUAL DRAINAGE BAG, ASSEMBLIES, AND RELATED METHODS, and U.S. Provisional Patent Application No. 62/938,452, filed Nov. 21, 2019, and titled DUAL DRAINAGE BAG, ASSEMBLIES, AND RELATED METHODS, each of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

The present disclosure relates generally to the field of medical devices. More specifically, the present disclosure relates to bodily fluid drainage assemblies and related methods of use.

BRIEF DESCRIPTION OF THE DRAWINGS

The embodiments disclosed herein will become more fully apparent from the following description and appended claims, taken in conjunction with the accompanying drawings. These drawings depict only typical embodiments, which will be described with additional specificity and detail through use of the accompanying drawings in which:

FIG. 1 is a plan view of a bodily fluid drainage assembly, depicted in an unassembled configuration.

FIG. 2 is a partial view of the bodily fluid drainage assembly of FIG. 1, depicted as it is being assembled.

FIG. 3 is a partial view of the bodily fluid drainage assembly of FIG. 1, depicted in an assembled configuration.

FIG. 4 is a partial view of the bodily fluid drainage assembly of FIG. 3, depicted with a hanger in a folded state.

FIG. 5A is a front view of the bodily fluid drainage assembly of FIG. 3, depicted with the hanger in an unfolded state.

FIG. 5B is a front view of the bodily fluid drainage assembly of FIG. 3, depicted with the hanger in partially folded state.

FIG. 5C is a front view of the bodily fluid drainage assembly of FIG. 3, depicted with the hanger in a folded state.

FIG. 6A is a front view of a bodily fluid drainage assembly, depicted with a cover.

FIG. 6B is a front view of the bodily fluid drainage assembly of FIG. 6A, depicted with the cover partially lifted.

FIG. 7 is an end view of the bodily fluid drainage assembly of FIG. 6A.

FIG. 8 is a front view of a bodily fluid drainage assembly mounted to a hospital bed rail.

FIG. 9 is a front view of a bodily fluid drainage assembly mounted to a hospital bed rail opposite of the hospital bed rail of FIG. 8.

FIG. 10 is a front view of a bodily fluid drainage assembly mounted to a rail or support structure.

FIG. 11A is a partial view of a bodily fluid drainage assembly, depicted with a hanger in an unfolded state.

FIG. 11B is a partial view of the bodily fluid drainage assembly of FIG. 11A, depicted with the hanger in a folded state.

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FIG. 12A is a partial view of a bodily fluid drainage assembly, depicted with a hanger in an unfolded state.

FIG. 12B is a partial view of the bodily fluid drainage assembly of FIG. 11A, depicted with the hanger in a folded state.

FIG. 13 is plan view of a bodily fluid drainage assembly, depicted in an unassembled configuration.

FIG. 14 is a perspective view of a forming structure that can be used with a bodily fluid drainage assembly.

FIG. 15 is a plan view of a bodily fluid drainage assembly, depicted in an unassembled configuration.

FIG. 16A is a partial view of a bodily fluid drainage assembly, depicted with a hanger in an unfolded state.

FIG. 16B is a partial view of the bodily fluid drainage assembly of FIG. 16A, depicted with the hanger in a folded state.

FIG. 17 is an end view of a bodily fluid drainage assembly.

FIG. 18A is a front view of a bodily fluid drainage assembly, depicted with the outlet tube in a folded configuration.

FIG. 18B is a front view of the bodily fluid drainage assembly of FIG. 18A, depicted with the outlet tube in an extended configuration.

FIG. 19 is a front view of another bodily fluid drainage assembly, depicted with the outlet tube in an extended configuration.

FIG. 20A is a partial view of a bodily fluid drainage assembly, depicted with a hanger in an unfolded state.

FIG. 20B is a partial view of the bodily fluid drainage assembly of FIG. 20A, depicted with the hanger in a folded state.

FIG. 21 is an exploded perspective view of the hanger and a base of the bodily fluid drainage assembly of FIG. 20A.

FIG. 22A is a top perspective view of the hanger and base of the bodily fluid drainage assembly of FIG. 20A, depicted in a pre-locked configuration.

FIG. 22B is a bottom perspective view of the hanger and base of the bodily fluid drainage assembly of FIG. 20A, depicted in the pre-locked configuration.

FIG. 23A is a top perspective view of the hanger and base of the bodily fluid drainage assembly of FIG. 20A, depicted in a locked configuration.

FIG. 23B is a bottom perspective view of the hanger and base of the bodily fluid drainage assembly of FIG. 20A, depicted in the locked configuration.

FIG. 24A is a front view of a cover of a bodily fluid drainage assembly.

FIG. 24B is a front view of another bodily fluid drainage assembly including the cover of FIG. 24A in an obscuring orientation.

FIG. 25 is a perspective view of another embodiment of a bodily fluid drainage assembly including a rotatable elbow connector.

DETAILED DESCRIPTION

A fluid drainage assembly may include a fluid bag configured to receive bodily fluid drained from a site within a human body, such as a bladder. The fluid bag may include various components, such as an inlet port configured to couple to a drainage tube, an outlet tube having an output regulator, one or more fluid compartments, one or more air vents (which can allow passage of gas or air into and/or out of the fluid compartments), and/or a volume indicator. In some instances, each fluid compartment may include transparent or translucent front and/or back panels. Each fluid

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compartment may also include a displaceable cover to obscure observation of the bodily fluid collected within the fluid compartment or fluid bag. In certain instances, the drainage assembly may include a transformable hanger coupled to the fluid bag and configured to suspend the fluid bag from a support structure, such as a bed rail. The hanger may be configured to transform configurations such that either the first fluid compartment or the second fluid compartment faces outward toward a clinician. In doing so, the fluid drainage assembly can be disposed at various locations on a support structure (e.g., bed or chair structure), and on either side of the support structure (e.g., bed or chair structure).

Embodiments may be understood by reference to the drawings, wherein like parts are designated by like numerals throughout. It will be readily understood by one of ordinary skill in the art having the benefit of this disclosure that the components of the embodiments, as generally described and illustrated in the figures herein, could be arranged and designed in a wide variety of different configurations. Thus, the following more detailed description of various embodiments, as represented in the figures, is not intended to limit the scope of the disclosure but is merely representative of various embodiments. While the various aspects of the embodiments are presented in drawings, the drawings are not necessarily drawn to scale unless specifically indicated.

It will be appreciated that various features are sometimes grouped together in a single embodiment, figure, or description thereof for the purpose of streamlining the disclosure. Many of these features may be used alone and/or in combination with one another.

The phrase “coupled to” refers to any form of interaction between two or more entities, including mechanical, electrical, magnetic, electromagnetic, fluid, and thermal interaction. Two components may be coupled to each other even though they are not in direct contact with each other. For example, two components may be coupled to each other through an intermediate component.

“Fluid” is used in its broadest sense, to refer to any fluid, including both liquids and gases as well as solutions, compounds, suspensions, etc., which generally behave as fluids.

FIG. 1 depicts a bodily fluid drainage assembly 100 in an unassembled configuration or state. The bodily fluid drainage assembly 100 includes a drainage bag assembly 110 and a hanger 150. The drainage bag assembly 110 includes one or more fluid compartments 121, 122. The fluid compartments 121, 122 can also be referred to as fluid bags or fluid receptacles. As detailed below, the drainage assembly 100 is configured to receive bodily drainage fluid; retain the drainage fluid; and/or selectively drain the drainage fluid from the drainage assembly 100. In some embodiments, the drainage assembly 100 is also configured to at least partially obscure the visibility of the drainage fluid while allowing an approximate volume of the fluid to be determined; and/or optionally allow the fluid to be directly viewed by at least partially displacing or removing a portion of the drainage assembly 100.

In the embodiment of FIG. 1, the drainage bag assembly 110 includes a first fluid compartment 121 and a second fluid compartment 122. The fluid compartments 121, 122 are in fluid communication with one another, such that fluid from one compartment 121, 122 can flow to and from the other compartment 121, 122. The flow can be unobstructed and free of interference. For instance, fluid from the first fluid compartment 121 can flow to the second compartment 122 and vice versa. At least one of the fluid compartments 121, 122 comprises an inlet port 116. At least one of the fluid

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compartments 121, 122 can comprise an air vent 117. Further, in some embodiments, each of the fluid compartments 121, 122 comprises an air vent 117. The air vent 117 can allow for air and/or gas to flow into and/or out of the fluid compartments 121, 122. For example, air and/or gas can be released from the fluid compartment 121, 122 as it fills with bodily fluid. Air and/or gas can also flow into the fluid compartment 121, 122 as the bodily fluid is drained from the fluid compartment 121, 122.

As shown in FIG. 1, the fluid compartments 121, 122 may comprise a front or first panel 123 and a back or second panel 124, which may be made of one or more liquid impervious materials. For example, the front and back panels 123, 124 may be formed from polyvinyl chloride, polyurethane, vinyl, polymeric, or any other suitable material. At least a portion of the front and back panels 123, 124 can be substantially transparent or translucent such that a drainage fluid contained within the fluid compartments 121, 122 may be readily observed. The panels 123, 124 may be coupled or joined along a other suitable technique. When coupled together, the panels 123, 124 form a fillable void that may receive drainage fluid via a drainage tube 112 and a corresponding inlet port 116 that are located on an upper portion of at least one of the fluid compartments 121, 122. The panels 123, 124 can also be flexible rather than rigid, such that the fluid compartments 121, 122 can expand as they fill with drainage fluid.

It will further be appreciated that a variety of types and configurations of fluid bags can be utilized without departing from the scope and spirit of the present disclosure. For example, the fluid compartments 121, 122 may be manufactured using a one-piece method, wherein the fluid compartments 121, 122 comprise a single piece of plastic that is folded along one or more edges and, further, is sealed to itself or otherwise closed along one or more edges. A two-piece manufacturing method can also be used, wherein two separate panels are coupled together. Further, the shape and size of drainage assembly 100 is primarily for illustrative purposes and may vary. In the depicted embodiment, the drainage bag assembly 110 is configured in a horizontal or landscape configuration where the height of the bag is less the width of the bag. This configuration allows for the drainage bag assembly 110 to be hung from a support, such as a hospital bed rail, that may be located as low as approximately seven inches from a floor surface—as illustrated in FIGS. 8-9 where D₁ may be lowered to less than 18 inches, less than 12 inches, less than 10 inches, or less than 8 inches (e.g., when the bed is in the lowered state).

As further shown in FIG. 1, the first and second fluid compartments 121, 122 can be defined by one or more seams 125, 126 that extend across a central region of the drainage bag assembly 110. For instance, the illustrated embodiment includes a first seam 125 and a second seam 126 that are disposed between the first and second fluid compartments 121, 122. The seams can be made in various ways, such as via radio frequency (RF) welding, heat sealing, gluing, or any other suitable technique. The seams 125, 126 can also be discontinuous, such that they do not block the flow of fluid between the fluid compartments 121, 122. For example, the seams 125, 126 can be referred to as spot seams (or spot welds).

The seams 125, 126 can couple one or more portions of the panels 123, 124 together. The seams 125, 126 can also define fold regions along which the drainage bag assembly 110 folds when assembled. In other embodiments, the first and second fluid compartments 121, 122 are defined by fold lines, as the drainage bag assembly 110 can be folded to

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form two halves or fluid compartments **121**, **122** (which can double the volume of fluid for a given bag height).

In certain embodiments, the drainage bag assembly **110** further includes one or more conduits **130** that can be configured to communicate with each of the fluid compartments **121**, **122**. In such embodiments, fluid can flow through the conduits **130** and between the fluid compartments **121**, **122**. The conduits **130** can be disposed within the drainage bag assembly **110**, and can be held in place via one or more seams **128**, which can be formed in various ways, such as via radio frequency (RF) welding, heat sealing, gluing, or any other suitable technique. The conduits **130** can also be held in place by an adhesive, such as a glue. In other embodiments, conduits **130** are not used.

As illustrated, an outlet tube **113** can also be disposed between the fluid compartments **121**, **122**. The outlet tube **113** may allow the drainage fluid contained within the fluid compartments **121**, **122** to be selectively drained from or retained within the fluid compartments **121**, **122** via an output regulator **114**. The output regulator **114** may comprise a plastic or metal clip, in-line valve, rotatable valve, or any other suitable structure. At least one of the fluid compartments **121**, **122** may further comprise an outlet tube holder **115** that comprises a slot, loop, pocket, or hook that is configured to receive and reversibly retain outlet tube **113** in an at least partially upright or folded position. For example, in the illustrated embodiment, the outlet tube holder **115** comprises a plastic tube so as to define a cavity. When the drainage assembly **100** is in use or a pre-use state, a bottom end of the outlet tube **113** can be positioned within the cavity of the outlet tube holder **115**. The outlet tube **113** can be removed from the outlet tube holder **115** and allowed to hang downwardly into the extended position shown in FIG. 7 during drainage of fluid from the fluid compartments **121**, **122**.

The drainage bag assembly **110** can be folded and/or assembled prior to use. For example, as shown in FIGS. 2-4, the drainage bag assembly **110** can be folded such that the first fluid compartment **121** and the second fluid compartment **122** are parallel to one another, or side by side. In some embodiments, the drainage bag assembly **110** is folded along seams **125**, **126** or predetermined fold lines. The first fluid compartment **121** can also be coupled to the second fluid compartment **122** such that the drainage bag assembly **110** can remain in the folded or assembled configuration.

For instance, as shown in FIG. 2, the drainage bag assembly **110** includes a base or brace **140** that extends between the first and second fluid compartments **121**, **122**. The brace **140** can be coupled to an elongate member **151** of each of the first and second fluid compartments **121**, **122**. For instance, with reference to FIGS. 2-4, each of the first and second fluid compartments **121**, **122** includes an elongate member **151**. In some embodiments, the elongate member **151** may be configured as a non-linear, elongate bar. In other embodiments, the elongate member **151** may be configured as a linear, elongate bar. In some instances, the elongate member **151** can also include horizontal and angled segments that permit the elongate member **151** to avoid interference with features of the drainage bag assembly **110**, such as the inlet port **116** and the air vent **117** when used with or coupled to the fluid compartments **121**, **122**. For example, a portion **159** of the elongate member **151** may extend toward the inlet port **116** and may include two horizontal segments and two angled segments. As another example, another portion **160** of the elongate member **151** may extend toward the air vent **117** and may include two horizontal segments and an angled segment disposed between the

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horizontal segments. The elongate member **151** may also include at least one retention feature **162**. The retention feature **162** may be configured to retain the wings **154**, **155** of the hanger **150** in a folded or deployed state as illustrated in FIG. 4. The retention feature **162** may be in the form of a bump, ridge, nub, boss, etc. or any other suitable form.

In FIG. 2, the brace **140** is shown coupled to the first elongate member **151a** and uncoupled from the second elongate member **151b**. The brace **140** can be coupled to each of the elongate members **151a**, **151b** in an analogous manner, or in a different manner, as desired. For instance, the brace **140** can include one or more coupling members **142** that can couple to each of the elongate members **151**. Thus, while only one side of the brace **140** is depicted in the illustrated embodiment, it will be appreciated that the opposite side of the brace **140** can be configured in a similar manner if desired.

The coupling members **142** can be configured to couple to the elongate member **151** in various ways. For instance, in the illustrated embodiment, the elongate member **151** includes complementary coupling members **143** that are configured to mate with or otherwise couple with the coupling members **142** of the brace **140**. In some instances, the coupling of the brace **140** to the elongate member **151** is non-reversible, such that the elongate members **151** (and/or fluid compartments **121**, **122**) are not intended to be unassembled or uncoupled from the brace **140** after being coupled thereto. With reference to FIG. 2, for instance, the brace **140** includes first coupling members **142** that are configured to be inserted into and/or received by second coupling members **143** of the second elongate member **151b**. Once inserted, the coupling members **142**, **143** can be formed such that they are not intended to be uncoupled (e.g., such as in a snap fit engagement). Various types of coupling members **142**, **143** can be used, including mating coupling members, latches, buttons, snap-fit engagements, and the like. In other embodiments, the brace **140** can be reversibly coupled to the one or more elongate members **151**.

FIGS. 3 and 4 depict the drainage bag assembly **110** in an assembled configuration. As shown in FIG. 3, the coupling members **142** of the brace **140** have been inserted into the coupling members of the elongate member **151**. The drainage bag assembly **110** further includes a hanger or rail clip **150**. The hanger or rail clip **150** can be coupled to or disposed on the brace **140**. For example, in the illustrated embodiment of FIGS. 3 and 4, the hanger **150** can be positioned and extend from the brace **140**.

The hanger **150** can be configured to allow the drainage assembly **100** to be suspended from a patient, or from a support structure, such as a wheelchair, bed, a bed rail, or stand. As shown in FIG. 4, the hanger can be positioned such that it is directed towards either side of the drainage bag assembly **110**. For instance, the hanger **150b** can be positioned towards the first fluid compartment **121**, or the hanger **150a** can be positioned towards the second fluid compartment **122**, as desired by the practitioner (e.g., a practitioner may suspend the drainage assembly **100** in either direction, depending on a desired orientation of the inlet port **116** and tube **112**.)

As illustrated in FIGS. 5A-5C, the hanger **150** may include a central portion **153**, a first laterally extending member or wing **154**, and a second laterally extending member or wing **155**. The hanger **150** may be formed as an integral unit using any suitable manufacturing technique, such as injection molding, casting, machining, etc. In other embodiments, the hanger **150** may be assembled from separate components using any suitable assembly technique.

The hanger **150** may be formed from any suitable material, such as a polymeric material like polypropylene, polyethylene, blends thereof, etc.

The central portion **153** may be coupled to the brace **140** and may also include a flexible or living hinge **158** such that a face of the central portion **153** is oriented parallel to a vertical axis or plane of the elongate members **151** in an undeployed state, as illustrated in FIG. 5A. The living hinge **158** may allow the central portion to pivot or fold about the living hinge **158** approximately or at least 180 degrees relative to the vertical axis or plane of the elongate member **151** (e.g., at least 90 degrees in either direction). In other words, the central portion **153** may be pivoted from a vertical orientation to a horizontal orientation on either side of the elongate member **151** when the hanger **150** is deployed, as depicted in FIGS. 5B-5C.

The first and second wings **154**, **155** may be coupled to the central portion **153** via a flexible or living hinge **156** such that the wings **154**, **155** are oriented in a plane of the face of the central portion **153** in the undeployed state, as shown in FIGS. 5B-5C. The living hinge **156** may allow the wings **154**, **155** to pivot or fold about the living hinge **156** approximately or at least 180 degrees relative to the plane of the face of the central portion **153** (e.g., at least 90 degrees in either direction). In other words, the wings **154**, **155** may be pivoted from the parallel orientation to a perpendicular orientation on either side of the central portion **153** when the hanger **150** is deployed, as depicted in FIG. 4. The wings **154**, **155** may include a support receiver **152** configured to be clipped over or disposed over a portion of a hospital bed rail. In other embodiments, the support receiver **152** may be configured to be clipped over or disposed over any suitable support, such as a wheelchair, a stand, etc. In certain embodiments, adjacent edges of the elongate member **151**, central portion **153** and wings **154**, **155** may be radiused or double beveled to facilitate pivoting or folding.

As shown in the depicted embodiment, the wings **154**, **155** may also include an elongate member receiver or elongate member section receiver **161**. The elongate member receiver **161** may have a substantially U-shaped form where a portion of the elongate member **151** can be removably disposed within the elongate member receiver **161** when the hanger **150** is deployed. In some instances, the retention feature(s) **162** can also aid in retaining the wings **154**, **155** in a folded or deployed state. For example, the wings **154**, **155** can be forced over the retention feature(s) **162** as the hanger **150** is folded or deployed. The retention feature(s) **162** can then retain the wings **154**, **155** in the folded or deployed state absent application of an outside force to unfold the wings **154**, **155**.

In certain embodiments, the wings **154**, **155** may further include brace extension receiver **162**. The brace extension receiver **162** may also have a substantially U-shaped form where a portion of the brace extension **144** can be removably disposed within the brace extension receiver **162** when the hanger **150** is deployed. The brace extensions **144** can protrude from the brace **140** and can be configured to further aid in retaining and/or supporting the hanger when in a deployed or folded state. The brace extension **144** can also include retention features if desired.

As previously discussed, in the illustrated embodiment, the hanger or rail clip **150** is configured to be oriented either toward the first fluid compartment **121** or the second fluid compartment **122** when deployed. This dual orientation capability of the hanger **150** permits the drainage assembly **100** to be suspended from a support, such as a bed rail, with either the first fluid compartment **121** or the second fluid

compartment **122** directed outwardly toward a viewer. Additionally, the dual orientation capability allows the viewer to orient the drainage assembly **100** at various locations along a support structure (e.g., upwards (towards the head) or downwards (towards the feet) from the bladder of a patient) or on either side of the support structure (e.g., bed structure) and still have the drainage port **116** adjacent a patient's drainage site in order to route the drainage tube **112** easily to the drainage site.

In some embodiments, an approximate volume of drainage fluid within the fluid bag **111** may be ascertained via a bag volume indicator **132**, which may comprise marks or graduations **134**, **135** on at least one panel **124** of each of the fluid compartments **121**, **122**. In the depicted embodiment, graduations **134**, **135** denote various predetermined volumes and may be printed on each of the fluid compartments **121**, **122**. In another embodiment, the graduations **134**, **135** may comprise raised or recessed portions of the fluid compartments **121**, **122** that are formed during or after manufacturing of the each of the fluid compartments **121**, **122**. The graduations **134**, **135** may represent any predetermined measurement of volume, such as fluid ounces and/or milliliters, and may vary in their relative distributions accordingly. In some embodiments, the space between sequential graduations **134**, **135** may not be uniform in scale so as to account for a non-linear rate of rise in fluid level, such as may result from expansion of the fluid compartment **121**, **122** as the fluid compartment **121**, **122** fills with drainage fluid and/or from a transverse dimension of the fluid compartment **121**, **122** that varies with a height of the bag. In the depicted embodiment, the graduations **134**, **135** are located on a right portion (as seen from the viewer's perspective) of the fluid compartment **121**, **122**; however, in other embodiments, the graduations may be located at any suitable location on the fluid compartment **121**, **122**. In the illustrated embodiment, the volume indicator **132** also extends upwardly. Since at least a portion of the fluid compartment **121**, **122** is transparent or semitransparent (e.g., translucent), the top of the drainage fluid can be compared to the graduations **134**, **135** to thereby determine the volume of the fluid, or the approximate volume thereof.

As shown in FIGS. 6A and 6B, in certain embodiments, one or more (or both) of the fluid compartments **221** further comprises a cover **281** that can be formed from or otherwise include an opaque material (or at least partially opaque material capable of obscuring the appearance of a fluid positioned at one side thereof), and that may define a similar shape as the fluid compartment **221**. For example, the cover **281** may be formed from polyvinyl chloride, polyurethane, vinyl, polymeric, or any other suitable material. As depicted, the cover **281** is located adjacent to and is configured to obscure the fluid compartment **221**. A second cover (not depicted) can also be disposed on an opposite side of the bag and in front of the second fluid compartment. The covers **281** have an outer edge that may be at least partially aligned with an outer edge of the fluid compartments **221**. However, at least a portion of the outer edge of the covers **281** may extend outwardly beyond the outer edge of the fluid compartment **221**, if desired.

The cover **281** can be coupled or attached to the fluid compartment **221** in any suitable manner, such as RF welding, heat sealing, gluing, hardware, etc. In the illustrated embodiment, the cover **281** is fixedly attached to the fluid compartment **221** along a seam which extends along an upper edge of the cover **281**. The seam may provide an area or region of weakness along which the cover **281** may be

readily removed from the fluid compartment **221**, if desired. Other regions of weakness (e.g., perforations) may also be used.

The cover **281** may be said to be integral with the fluid compartment **221**. For example, in some embodiments, the drainage assembly **200** is sterilized and packaged with all of the components shown in FIG. 6A. While it may be possible to remove the cover **281** from the drainage assembly **200**, as just discussed, the cover **281** is nevertheless a fixed and permanent component of the drainage assembly **200**. For example, if the cover **281** is removed by peeling it along its region of weakness, a portion of the cover **281** (generally the portion that is attached to the panels **221** by the seam) can nevertheless remain attached to the fluid compartment **221**. In other embodiments, the cover **281** can be selectively attachable to and detachable from the drainage assembly **200**, such as with snaps, hook-and-loop fasteners, adhesives, or any other suitable temporary fastening system. For example, the fasteners can be positioned along an upper edge of the cover **281**.

As previously mentioned, the cover **281** is coupled or attached to the drainage assembly **200** at an upper end of the fluid compartment **221**. The cover **281** thus naturally hangs or drapes downward so as to be positioned in front of the face of the panel **224** and/or fluid compartment **221**. The cover **281** can thus conceal or obscure viewing of a large portion of the panel **224** and/or fluid compartment **221**, when the drainage assembly **200** is viewed from a position in front of or behind the drainage assembly **200**. A lower edge of the cover **281** may not be attached to the fluid compartment **221** such that at least a portion of the cover **281** may be lifted to expose the fluid compartment **221** and its contents. In other embodiments, the lower edge of the cover **281** may be selectively attached to the fluid compartment **221**. Such selective attachment and detachment may be achieved in any suitable manner, such as via snaps, hook-and-loop fasteners, adhesives, or other suitable releasable fastening system. In other embodiments, a cover can be configured as a slip cover that is disposed over the fluid bag assembly. For instance, a polymeric or fabric cover can be disposed over the fluid bag assembly rather than coupled thereto.

The cover **281** can include one or more volume indicators **285**. The illustrated first volume indicator **285** includes a window **282** and one or more graduations **283**. A variety of configurations for the window **282** and the graduations **283** are possible. For example, a variety of shapes are possible for the window **282**, and the number and arrangement of the graduations **283** may be varied. In the illustrated embodiment, the window **282** defines a permanent opening through the cover **281** and is fully encompassed by solid or unbroken portions of the cover **281**. The window **282** defines a small, vertically oriented longitudinal slot. Only a small portion of the fluid compartment **221** is visible through the window **282** when the cover **281** are in the obscuring orientation. An approximate volume level can be determined when a top level of the drainage fluid is visible through the window **282**, such as by comparing the level of the drainage fluid to the graduations **283**. Due to the fixed relationship between the upper end of the cover **281** and the fluid compartment **221**, the graduations **283** can provide a substantially accurate assessment of the approximate liquid volume when the cover **281** hangs downwardly.

In the illustrated embodiment, the window **282** may be oriented such that a portion of the graduations **234**, **235** that are on the fluid compartment **221** are visible through the window **282** when the cover **281** are in the obscuring orientation. In certain embodiments, the bag graduations

234, **235** may be offset relative to the cover graduations **283** when the fluid compartment **221** is devoid of fluid. The offset can account for relative movement between the cover **281** and the fluid compartment **221** as the fluid compartment **221** fills with fluid. In particular, the fluid compartment **221** can transition from a generally flat configuration to an outwardly expanded or bowed configuration as it fills with the drainage fluid, and in some arrangements, the cover **281** may not strictly match the contour of the fluid compartment **281** as it expands. This relative shift in positions can cause the graduations **283** of the first volume indicator **285** to come into greater alignment with the graduations **234**, **235** of the bag volume indicator **232**.

The cover **281** may include a second volume indicator **286**. The second volume indicator **286** includes a recess, such as a notch or cutout **287**, and a set of graduations **288**. A bottom end of the cutout **287** is at a position that is horizontally offset and/or spaced vertically from an upper end of the window **282**. Due to the difference in vertical positions between the cutout **287** and the window **282**, a top level of the drainage fluid can remain below the cutout **287** at all times that the window **282** is used in determining the approximate volume of the drainage fluid, and the top level of the drainage fluid can remain above the window **282** at all times that the cutout **287** is used in determining the approximate volume of the drainage fluid.

Each of the window **282** and the cutout **287** may be referred to as an aperture or opening in the cover **281** through which a portion of the panel **224** of the fluid compartment **221** may be directly viewed. The window **282** and the cutout **287** may be formed in any suitable manner, such as, for example, by stamping, cutting, or punching.

By way of further explanation, in the illustrated embodiment, the cover **281** is formed of a flexible material. When the drainage assembly **200** is in the unfilled state, the fluid compartment **221** can be substantially planar or flat, and the cover **281** can hang substantially vertically and may likewise be substantially planar or flat. The cover **281** may be positioned exclusively forward of the panel **224**, and it may obscure an entirety of the panel **224** or fluid compartment **221** (except for those portions that are visible through the window **282** and the cutout **287**) from all or nearly all vantage points that are forward of the drainage assembly **200**. A second cover (not depicted) may be positioned exclusively behind the second fluid compartment, and it may obscure an entirety of the second fluid compartment (except for those portions that are visible through a window and a cutout) from all or nearly all vantage points that are rearward or backward of the drainage assembly **200**.

In FIG. 6A, the cover **281** is shown in an obscuring (e.g., draped or lowered) orientation. In other words, the cover **281** is shown obscuring at least a portion of the panel **224** and/or fluid compartment **221**. In FIG. 6B, the cover **281** is shown in a viewing (e.g., raised) orientation. In other words, the lower edge of the cover **281** is shown lifted to reveal at least a portion of the panel **224** and/or fluid compartment **221**. A second cover can be configured to function in a similar manner relative to the second fluid compartment. It is noted that the term "obscuring orientation," when used with respect to the cover **281**, does not necessarily connote that an entirety of the panel **224** and/or fluid compartment **221** is obscured (e.g., blocked or guarded from view, rendered difficult to view, or provided with a significantly altered appearance) by the cover **281**. Indeed, as demonstrated by the embodiment depicted in FIG. 6A, restricted portions of the panel **224** and/or fluid compartment **221** can

be viewed, observed, or visualized through the window 282 and the cutout 287 when the cover 281 is in the obscuring orientation.

FIG. 7 depicts an end view of a drainage assembly 200 that includes first and second fluid compartments 221, 222 each having a cover 281, 284 coupled thereto. As shown in FIG. 7, the covers 281, 284 drape over the outwardly facing surfaces of the first and second fluid compartments 221, 222. In such a manner, fluid 205 within the fluid compartments 221, 222 can be obscured from view by an observer. A bottom portion of the covers 281, 284 can also be lifted as desired to view the fluid 205 within the fluid compartments 221, 222.

The drainage assembly 200 further includes a hanger 250 that is configured to couple to a rail 202 or other structure. In the illustrated embodiment, the hanger 250 is depicted in a folded configuration or state, wherein the support receiver 262 is disposed over a rail 202. Additionally, the elongate member receiver 261 is depicted as being disposed over an elongate member 251 of the first fluid compartment 221, and the brace extension receiver 162 is depicted as being disposed over a brace extension 244 of the brace 240.

Additionally, FIG. 7 depicts a conduit 230 in communication with the first and second fluid compartments 221, 222. Fluid 205 can flow through a lumen 232 in the conduit 230 such that the fluid level in each of the fluid compartments is substantially the same. The conduit 230 also includes a notch or opening 231 such that fluid can flow out of the conduit 230 and towards the drainage tube 213. One or more conduits 230 can be used. Other structures can also be used to ensure flow is not substantially restricted (and can flow freely) between the fluid compartments 221, 222.

FIGS. 8-9 depict the drainage assembly 200 in use with a patient 203. As shown therein, the drainage assembly 200 may be suspended from a bed rail 202 that is coupled to a hospital bed 201. As discussed previously, the bed rail 202 may be positioned adjacent the floor such that a distance from the bed rail 202 to the floor is distance D_1 . In some embodiments, as depicted in FIG. 8, the hospital bed 201 is configured to be lowered to be adjacent the floor (which can minimize a fall distance if a patient 203 falls out of the hospital bed 201). For instance, D_1 may be lowered to less than 18 inches, less than 12 inches, less than 10 inches, or less than 8 inches (e.g., when the bed 201 is in the lowered state). Stated another way, the bed 201 may be lowered such that D_1 is between about 6 and about 18 inches, between about 6 and about 12 inches, between about 6 and about 10 inches, or between about 6 and about 8 inches. In such embodiments, the drainage assembly 200 can be suspended from the bed rail 202 such that it does not touch the floor, even while the hospital bed 201 is in the lowered state. The hanger 250 can be folded to permit the hospital bed 201 to be lowered closer to the floor because of the low-profile configuration of the deployed hanger 250.

The drainage assembly 200 may also be oriented on either side of the bed structure 201 (or at various locations along the bed rail 202 (e.g., more towards the head or more towards the feet of the patient 203), such that either the first fluid compartment 221 and the first cover 281 are directed outwardly as shown in FIG. 8 (which depicts a first side of the hospital bed 201 or bed rail 102) or such that the second fluid compartment 222 and the second cover 284 are directed outwardly as shown in FIG. 9 (which depicts a second or opposite side of the hospital bed 201 or bed rail 202). For example, the hanger or rail clip 250 may be deployed such that the hanger 250 is oriented toward the second fluid compartment 222 when the first fluid compart-

ment 221 is directed outwardly. In such instances, the central portion 253 may be pivoted or folded towards the second fluid compartment 222 and the wings 254, 255 may be pivoted or folded downwardly toward the base region 251 of the hanger 250 such that a portion of the elongate member 251 is received within the elongate member receiver 261 (e.g., shown in phantom in FIG. 10). If the user desires that the second fluid compartment 222 and the second cover 284 be directed outwardly, the hanger 250 may be deployed such that the hanger 250 is oriented toward the first fluid compartment 221. The central portion 253 may be pivoted or folded towards the first fluid compartment 221 and the wings 254, 255 may be pivoted or folded downwardly toward the elongate member 251 of the hanger 250 such that a portion of the base region 251 is received within the elongate member receiver 262 (e.g., shown in FIG. 10).

The drainage tube 212 may also be coupled to the patient's drainage site such that drainage fluid can fill the fluid compartments 221, 222. The first cover 281 and/or the second cover 284 may obscure the drainage fluid within the fluid compartments 221, 222. A viewer may observe the volume of drainage fluid within the fluid compartments 221, 222 by viewing the drainage fluid through the first volume indicator 285 and/or the second volume indicator 286. Alternatively, the healthcare worker may observe the volume of drainage fluid within a fluid compartment 221, 222 by lifting a lower portion of the outwardly directed cover 281, 284 and observing the drainage fluid volume through the outwardly directed fluid compartment 221, 222.

Various other types of hangers can also be used in accordance with the fluid assemblies, as shown in FIGS. 11A-11B and 12A-12B. For instance, the hanger 350 of FIGS. 11A-11B does not include a brace receiver or slot. Rather, the hanger is only coupled to the elongate member 351. And in FIGS. 12A-12B, the hanger 450 does not couple to the brace or the elongate member. Other types of hangers can also be used.

FIG. 13 depicts a bodily fluid drainage assembly 500 in accordance with another embodiment. As shown in FIG. 13, the bodily fluid drainage assembly 500 includes a forming structure 529 that can aid in allowing fluid to flow between the first and second fluid compartments 521, 522. The forming structure 529 can be disposed within the bag structure. For instance, the forming structure 529 can be disposed between the first and second fluid compartments 521, 522 and can be held in place via one or more seams 525, 526. The forming structure 529 can provide a form or shape to a bottom portion of the bag structure (e.g., after it is folded and/or assembled to form first and second fluid compartments 521, 522). The height and/or size of the forming structure can vary as desired. A perspective view of an exemplary forming structure 529 is shown in FIG. 14. Other types and/or shapes and sizes of forming structures can also be used.

FIG. 15 depicts another embodiment of a bodily fluid drainage assembly 600 in an unassembled configuration or state. The bodily fluid drainage assembly 600 includes a drainage bag assembly 610 and a hanger 650. In the embodiment of FIG. 15, the drainage bag assembly 610 includes a first fluid compartment 621 and a second fluid compartment 622. The fluid compartments 621, 622 are in fluid communication with one another, such that fluid from one compartment 621, 622 can flow to and from the other compartment 621, 622. At least one of the fluid compartments 621, 622 comprises an inlet port 616. If desired, at least one of the fluid compartments 621, 622 can comprise an air vent (such as the air vent 117 depicted in FIG. 1).

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As shown in FIG. 15, the fluid compartments 621, 622 may comprise a front or first panel 623 and a back or second panel 624, which may be made of one or more liquid impervious materials. For example, the front and back panels 623, 624 may be formed from polyvinyl chloride, polyurethane, vinyl, polymeric, or any other suitable material. At least a portion of the front and back panels 623, 624 can be substantially transparent or translucent such that a drainage fluid contained within the fluid compartments 621, 622 may be readily observed. The panels 623, 624 may be coupled or joined along a other suitable technique. When coupled together, the panels 623, 624 form a fillable void that may receive drainage fluid via a drainage tube 612 and a corresponding inlet port 616 that are located on an upper portion of at least one of the fluid compartments 621, 622. The panels 623, 624 can also be flexible rather than rigid, such that the fluid compartments 621, 622 can expand as they fill with drainage fluid.

As further shown in FIG. 15, the first and second fluid compartments 621, 622 can be defined by one or more seams 625, 626 that extend across a central region of the drainage bag assembly 610. For instance, the illustrated embodiment includes a first seam 625 and a second seam 626 that are disposed between the first and second fluid compartments 621, 622. The seams 625, 626 can couple one or more regions of the panels 623, 624 together. The seams 625, 626 can also define fold lines or regions where the drainage bag assembly 610 can fold when assembled.

The seams can be made in various ways, such as via radio frequency (RF) welding, heat sealing, gluing, or any other suitable technique. The seams 625, 626 can also be discontinuous, such that they do not block the flow of fluid between the fluid compartments 621, 622. For example, the seams 625, 626 can be referred to as spot seams (or spot welds). In some embodiments, each seam 625, 626 can comprise a series of spot seams (or spot welds). For instance, each seam 625, 626 can comprise three, four, five, six, seven, eight or more spot seams (or spot welds) along a distance of between about 8 and about 16 inches. In particular embodiments, each seam 625, 626 can comprise four, five, or six spot seams (or spot welds) along a distance of between about 8 and about 16 inches. And in more particular embodiments, each seam 625, 626 can comprise five or six spot seams (or spot welds) along a distance of between about 8 and about 16 inches. In some of such embodiments, the drainage bag assembly 610 does not include other conduits (such as conduits 130 in FIG. 1). In such embodiments, fluid can freely flow through the seamed regions 625, 626 and between the first and second fluid compartments 621, 622.

As illustrated, an outlet tube 613 can also be disposed between the fluid compartments 621, 622. The outlet tube 613 may allow the drainage fluid contained within the fluid compartments 621, 622 to be selectively drained from or retained within the fluid compartments 621, 622 via an output regulator 614. The output regulator 614 may comprise a plastic or metal clip, in-line valve, rotatable valve, or any other suitable structure. At least one of the fluid compartments 621, 622 may further comprise an outlet tube holder 615 that comprises a slot, loop, pocket, or hook that is configured to receive and reversibly retain outlet tube 613 in an at least partially upright or folded position. For example, in the illustrated embodiment, the outlet tube holder 615 comprises a plastic tube so as to define a cavity. When the drainage assembly 600 is in a packaged, folded, or filling configuration, a bottom end of the outlet tube 613 can be positioned within the cavity of the outlet tube holder 615. The outlet tube 613 can be removed from the outlet tube

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holder 615 and allowed to hang downwardly into an extended position shown in FIG. 18B during drainage of fluid from the fluid compartments 621, 622.

As further shown in the illustrated embodiment, the fluid compartments 621, 622 can comprise extension members 672, 674 that couple to an extended outlet tube section 670. For example, in FIG. 15 the fluid compartments 621, 622 each comprise an extension member 672, 674 that extends outwardly from the fluid compartment 621, 622. The extension members further couple to the outlet tube section 670 which extends from a central region of the drainage bag assembly 610. The outlet tube section 670 is also coupled to the outlet tube 613 to allow drainage from the drainage bag assembly 610.

In some embodiments, the extension members 672, 674 are integral with the fluid compartments 621, 622. For instance, the extension members 672, 674 can be formed of the same panel material. The extension members 672, 674 can also be defined by one or more seams 676, 677, 678, 679. If desired, the seams 676, 677, 678, 679 can be configured to not allow the passage of fluid. In other embodiments, fluid can be allowed to flow into the extension members 672, 674 and to the outlet tube section 670. When the drainage bag assembly 610 is folded into an assembled state, the extension members 672, 674 can also provide support for the outlet tube section 670. For instance, the extension members 672, 674 can be coupled and configured to retain the outlet tube section 670 in an extended configuration, as further discussed below in relation to FIG. 18B.

FIGS. 16A and 16B depict a portion of the drainage bag assembly 610 in the folded and/or assembled configuration. As shown therein, the drainage bag assembly 610 is folded such that the first fluid compartment 621 and the second fluid compartment 622 are parallel to one another, or side by side. The first fluid compartment 621 is also coupled to the second fluid compartment 622 such that the drainage bag assembly 610 can remain in the folded or assembled configuration during use.

In the illustrated embodiment, the drainage bag assembly 610 includes a base or brace 640 that extends between the first and second fluid compartments 621, 622. For instance, the brace 640 can be coupled to an elongate member 651 of each of the first and second fluid compartments 621, 622 to retain the drainage bag assembly 610 in the assembled configuration. A hanger 650 is also coupled to and extending upward from the base 640.

The hanger 650 can be configured to allow the drainage assembly 600 to be suspended from a patient, or from a support structure, such as a wheelchair, bed, a bed rail, or stand. As shown in FIGS. 16A and 16B, the hanger 650 can be positioned such that it is directed towards either side of the drainage bag assembly 610. For instance, the hanger 650a can be positioned towards the first fluid compartment 621, or the hanger 650b can be positioned towards the second fluid compartment 622, as desired by the practitioner (e.g., a practitioner may suspend the drainage assembly 600 in either direction, depending on a desired orientation of the inlet port 616 and tube 612). The hanger 650 may be formed as an integral unit using any suitable manufacturing technique, such as injection molding, casting, machining, etc. In other embodiments, the hanger 650 may be assembled from separate components using any suitable assembly technique. The hanger 650 may be formed from any suitable material, such as a polymeric material like polypropylene, polyethylene, blends thereof, etc.

As further illustrated in FIGS. 16A-16B, the hanger 650 may include a central portion 653, a first extending member

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or arm 654, and a second extending member or arm 655. The central portion 653 may be coupled to the brace 640 and may also include a flexible or living hinge 658 such that a face of the central portion 653 can be configured to flex and/or bend towards the first and second fluid compartment 621, 622 if desired. For instance, the hanger 650 can be bent towards either the first or second fluid compartment 621, 622 during packaging and/or shipping. The hanger 650 can also be bent towards either the first or second fluid compartment 621, 622 when being suspended from an object.

The first and second arms 654, 655 can also be coupled to the central portion 653 via a flexible or living hinge 656 such that the arms 654, 655 are oriented in a plane of the face of the central portion 653 in the undeployed state, as shown in FIG. 16A. The living hinge 656 may allow the arms 654, 655 to pivot or fold about the living hinge 656 approximately or at least 180 degrees relative to the plane of the face of the central portion 653 (e.g., at least 90 degrees in either direction). In other words, the arms 654, 655 may be pivoted from the parallel orientation to a perpendicular orientation on either side of the central portion 653 when the hanger 650 is deployed, as depicted in FIG. 16B. The arms 654, 655 may include a support receiver 652 configured to be clipped over or disposed over a portion of a hospital bed rail. In other embodiments, the support receiver 652 may be configured to be clipped over or disposed over any suitable support, such as a wheelchair, a stand, etc.

Although not depicted in FIGS. 16A and 16B, it will be appreciated that the drainage bag assembly 610 can further include one more covers (similar to covers 281, 284 shown in FIG. 7 or cover 981 shown in FIG. 24A). The drainage bag assembly 610 can further include one or more other features previously discussed, such as one or more graduations (like graduations 134, 135 shown in FIGS. 5A-5C), etc.

FIG. 17 depicts an end view of a drainage assembly 600 that includes first and second fluid compartments 621, 622 each having a cover 681, 684 coupled thereto. As shown in FIG. 17, the covers 681, 684 can couple to an upper region of the first and second fluid compartments 621, 622, and drape downwardly over the outwardly facing surfaces of the fluid compartments 621, 622. In such a manner, fluid 605 within the fluid compartments 621, 622 can be obscured from view by an observer. As previously discussed, a bottom portion of each cover 681, 684 can also be lifted as desired to view the fluid 605 within the fluid compartments 621, 622.

FIG. 17 also depicts a hanger 650 extending from a base 640 that couples the fluid compartments 621, 622 together. As shown in FIG. 17, the hanger 650 is configured to couple to a rail 602 or other structure. In the illustrated embodiment, the hanger 650 is depicted in a folded configuration or state, wherein each arm 654, 655 is pivoted or folded about a hinge 656 towards the first fluid compartment 621. The support receiver 652 is also shown disposed over a rail 602. If desired, the arms 654, 655 could also pivot or fold towards the second fluid compartment 622 (e.g., in instances where the rail is on the other side of the drainage bag assembly 610). The hanger 650 can also pivot or fold about hinge 658 if required.

As further shown in FIG. 17, the fluid 605 height is substantially the same in each of the first and second fluid compartments 621, 622 as the fluid can flow freely between the fluid compartments 621, 622. Prior to draining fluid 605 from the drainage bag assembly 610, the extension members 672, 674 and outlet tube section 670 are folded inwards

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towards the drainage bag assembly 610 and the outlet tube 613 is disposed in or housed within the outlet tube holder 615.

FIGS. 18A and 18B depict the drainage assembly 600 in a folded or assembled configuration during use. Specifically, FIG. 18A depicts the drainage assembly 600 with the outlet tube 613 in a folded configuration, and FIG. 18B depicts the drainage assembly 600 with the outlet tube 613 in an extended configuration. When folded, the fluid compartment 622 is defined at a bottom end by seam 625. The drainage bag assembly 610 is also shown without covers 681, 684 for ease of reference, but it is understood that one or more covers 681, 684 could be coupled to the drainage assembly 600 if desired.

With reference to FIG. 18A, the outlet tube 613 can be disposed in an upright or folded configuration prior to draining fluid from the drainage bag assembly 610. For instance, the outlet tube 613 can be disposed in the folded configuration while the drainage bag assembly 610 is being filled with fluid. In the folded configuration, the outlet tube is positioned or housed within the outlet tube holder 615. The extension members 672, 674 are also folded inwardly towards the drainage bag assembly 610.

When a practitioner desires to empty or drain fluid from the drainage bag assembly 610, the outlet tube 613 can be transitioned to an extended or drainage configuration, as shown in FIG. 18B. When transitioning to the extended configuration, the outlet tube 613 is removed from the outlet tube holder 615, pulled outwards away from the drainage bag assembly 610, and pivoted downwards (e.g., between about 150-200 degrees, or about 180 degrees). In doing so, the outlet tube section 670 is also pivoted from a first position disposed inwards towards the drainage bag assembly 610 to a second position disposed outwards and extending from the drainage bag assembly 610. In the extended or drainage configuration, the outlet tube 613 is also directed downwards for drainage.

When transitioning to the extended configuration, the extension members 672, 674 are also inverted and folded from a first position disposed inwards towards the drainage bag assembly 610 to a second position disposed outwards and extending from the drainage bag assembly 610. In the extended position, the extension members 672, 674 can support the outlet tube section 670 to help retain the outlet tube 613 in the downward orientation. The extension members 672, 674 can also aid in keeping the outlet tube section 670 from kinking or bending which can restrict flow to the outlet tube 613.

After a practitioner empties or drains fluid from the drainage bag assembly 610, the practitioner can transition the outlet tube 613 back to the folded configuration. It will thus be appreciated that the outlet tube 613 can be transitioned between the folded configuration of FIG. 18A and the extended configuration of FIG. 18B as desired.

FIG. 19 depicts a drainage assembly 700 according to yet another embodiment. In FIG. 19, the bottom section of the drainage bag assembly 710 is configured to angle and/or slope towards the outlet tube section 770 and outlet tube 713. For instance, each fluid compartment 721 can be configured such that the seam 725 defines an angle or slope when the drainage bag assembly 710 is folded and assembled for use. As shown in FIG. 19, the height H2 on the first side or end of the fluid compartment 722 is greater than the height H1 on the second side or end such that the bottom end of the fluid compartment 722 is not parallel with the top end. Configuring the fluid compartments 721, 722 with a bottom that slopes and angles towards the extension

members 772, 774 and outlet tube section 770 can aid in directing the flow of fluid towards the outlet tube 713 for optimal drainage and emptying of the drainage bag assembly 710.

FIGS. 20A-23B depict another embodiment of a drainage bag assembly 810. FIGS. 20A and 20B depict a portion of the drainage bag assembly 810 in the folded and/or assembled configuration. As shown therein, the drainage bag assembly 810 is folded such that the first fluid compartment 821 and the second fluid compartment 822 are parallel to one another, or side by side. The first fluid compartment 821 is also coupled to the second fluid compartment 822 such that the drainage bag assembly 810 can remain in the folded or assembled configuration during use.

In the illustrated embodiment of FIG. 20A-21, the drainage bag assembly 810 includes a base or brace 840 that extends between the elongate members 851. The base 840 may include a first base portion 840a coupled to and extending laterally from one elongate member 851 and a second base portion 840b coupled to and extending laterally from another elongate member 851. The first base portion 840a and the second base portion 840b may be configured to fixedly couple together. For instance, the first and second base portions 840a, 840b may include passages 845 that engage with pins or protrusions 846 when the first and second base portions 840a, 840b are coupled together. Other coupling configurations, such as snap fit, adhesive, loop and hook, etc., are also contemplated within the scope of the disclosure. In some embodiments, the first and second base portions 840a, 840b may be configured to releasably couple together. The base 840 may further include a slot 847 configured for coupling of a hanger 850 to the base 840.

The hanger 850 may be coupled to and extend upward from the base 840. The hanger 850 can be configured to allow the drainage bag assembly 810 to be suspended from a patient, or from a support structure, such as a wheelchair, bed, a bed rail, or stand. As shown in FIG. 20B, the hanger 850 can be positioned such that it is directed towards either side of the drainage bag assembly 810. For instance, the hanger 850a can be positioned towards the first fluid compartment 821, or the hanger 850b can be positioned towards the second fluid compartment 822, as desired by the practitioner (e.g., a practitioner may suspend the drainage bag assembly 810 in either direction, depending on a desired orientation of the inlet port 816 and tube 812). The hanger 850 may be formed from any suitable material, such as a polymeric material like polypropylene, polyethylene, blends thereof, etc.

As further illustrated in FIGS. 20A-21, the hanger 850 may include a central portion 853, a first extending member or arm 854, a second extending member or arm 855, and a base coupler 863. The central portion 853 may be coupled to the base 840 and may also include a flexible or living hinge 858 such that a face of the central portion 853 can be configured to flex and/or bend towards the first and/or second fluid compartment 821, 822 if desired. For instance, the hanger 850 can be bent towards either the first or second fluid compartment 821, 822 during packaging and/or shipping. The hanger 850 can also be bent towards either the first or second fluid compartment 821, 822 when being suspended from an object.

The first and second arms 854, 855 can also be coupled to the central portion 853 via a flexible or living hinge 856 such that the arms 854, 855 are oriented in a plane of the face of the central portion 853 in the undeployed state, as shown in FIG. 20A. The living hinge 856 may allow the arms 854, 855 to pivot or fold about the living hinge 856 approximately or

at least 180 degrees relative to the plane of the face of the central portion 853 (e.g., at least 90 degrees in either direction). In other words, the arms 854, 855 may be pivoted from the parallel orientation to a perpendicular orientation on either side of the central portion 853 when the hanger 850 is deployed, as depicted in FIG. 20B. The arms 854, 855 may also include a support receiver 852 configured to be clipped over or disposed over a portion of a support structure (e.g., like a hospital bed rail).

The hanger 850 may be coupled to the base 840 using any suitable assembly technique. For instance, as depicted in FIGS. 21-23B, the base coupler 863 of the hanger 850 can be disposed within the slot 847 of the base 840 such that a cylindrical central portion 866 is disposed within the slot 847, an upper portion 857 is disposed above the slot 847, and a lower portion 859 is disposed below the slot 847. As depicted in FIGS. 22A and 22B, the hanger 850 is oriented perpendicular to a longitudinal axis of the elongate members 851 in a pre-locked configuration. As shown in FIG. 22B, the lower portion 859 is positioned at a base of ramps 864 which are disposed on a bottom surface of the first and second base portions 840a, 840b. The ramps 864 are inclined downwardly away from the bottom surface and define a channel 865.

FIGS. 23A-23B illustrate the hanger 850 in a locked configuration. In the locked configuration, the hanger has been rotated 90 degrees in a first direction from the pre-locked configuration such that it is oriented parallel to the longitudinal axis of the elongate members 851. During rotation, the lower portion 859 engages with the ramps 864 and is turned such that it moves to a position where it is disposed within the channel 865 at the end of the rotation. The channel 865 is disposed between the ramps 864 and is configured to restrict rotation of the hanger 850 in a second direction to decouple the hanger 850 from the base portions 840a, 840b. For instance, the ramps 864 can include shoulders that restrict rotation of the hanger 850 in the second or reverse direction. When in the locked configuration, the base portions 840a, 840b can also be locked in their engaged position such that they cannot be separated or otherwise pulled apart. The bag assembly 810 can thus remain in the assembled or folded configuration for use. In other embodiments, the hanger 850 can be rotated in a reverse direction (after being locked or engaged) by applying a sufficient force such that the hanger 850 can optionally be removed from the channel 865, and the base portions 840a, 840b can be optionally separated.

Although not depicted in FIGS. 20A and 20B, it will be appreciated that the drainage bag assembly 810 can further include one more covers (similar to covers 281, 284 shown in FIG. 7 or cover 981 shown in FIG. 24A). The drainage bag assembly 810 can further include one or more other features previously discussed, such as one or more graduations (like graduations 134, 135 shown in FIGS. 5A-5C), etc.

As shown in FIGS. 24A and 24B, in certain embodiments, a bodily fluid drainage assembly 900 can include a drainage bag assembly 910 comprising a cover 981 that can be formed from or otherwise include an opaque material (or at least partially opaque material capable of obscuring the appearance of a fluid within the drainage bag assembly 910 when in use), and that may define a sleeve to selectively cover portions of the drainage bag assembly 910. For example, the cover 981 may be formed from polyvinyl chloride, polyurethane, vinyl, polymeric, or any other suitable material. The cover 981 may comprise front and back panels 989 that define a cavity configured to receive the

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drainage bag assembly 910. Side ends or edges of the panels 989 may be welded together and lower ends or edges of the panels 989 may be uncoupled defining a lower opening 997 to the cavity. Upper ends or edges of the panels 989 may define an upper opening 996 and may include one or more fasteners 998 configured to selectively couple the upper ends together to close the upper opening 996. The fasteners 998 may be of any suitable type, such as snaps, clips, hook-and-loop, zippers, buttons, etc. As depicted in FIG. 24B, the cover 981 is located adjacent to and is configured to obscure a fluid compartment 921.

The cover 981 can include one or more volume indicators 985. The illustrated first volume indicator 985 includes a window 982 and one or more graduations 983. A variety of configurations for the window 982 and the graduations 983 are possible. For example, a variety of shapes are possible for the window 982, and the number and arrangement of the graduations 983 may be varied. In the illustrated embodiment, the window 982 defines a permanent opening through the cover 981 and is fully encompassed by solid or unbroken portions of the cover 981. The window 982 defines a small, vertically oriented longitudinal slot. Only a small portion of the fluid compartment 921 is visible through the window 982 when the cover 981 is in an obscuring orientation. An approximate volume level can be determined when a top level of the drainage fluid is visible through the window 982, such as by comparing the level of the drainage fluid to the graduations 983. Due to the fixed relationship between the upper end of the cover 981 and the fluid compartment 921, the graduations 983 can provide a substantially accurate assessment of the approximate liquid volume when the cover 981 is in the obscuring orientation.

The cover 981 may include a second volume indicator 986. The second volume indicator 986 includes a recess, such as a notch or cutout 987, and a set of graduations 988. A bottom end of the cutout 987 is at a position that is horizontally offset and/or spaced vertically from an upper end of the window 982. Due to the difference in vertical positions between the cutout 987 and the window 982, a top level of the drainage fluid can remain below the cutout 987 at all times that the window 982 is used in determining the approximate volume of the drainage fluid, and the top level of the drainage fluid can remain above the window 982 at all times that the cutout 987 is used in determining the approximate volume of the drainage fluid. In some embodiments, the cover 981 may include volume indicators similar to first and second volume indicators 985, 986 disposed on both the front and back panels 989 configured to determine the approximate volume of the drainage fluid within first and second drainage bags.

In FIG. 24B, the cover 981 is shown in the obscuring orientation. In other words, the cover 981 is shown disposed over the drainage bag assembly 910 and obscuring at least a portion of the fluid compartment 921. The cover 981 may be disposed in the obscuring orientation by unfastening the fasteners 998 to open the upper opening 996 of the cover 981 and sliding the drainage bag assembly 910 into the cavity of the cover 981 such that a hanger 950 extends at least partially through the upper opening 996 and an output regulator 914 and an output tube 913 extends at least partially through the lower opening 997. The hanger 950 may be similar to any of the previously disclosed hangers of the present disclosure. Alternatively, the drainage bag assembly 910 may be slid into the cavity through the lower opening 997 of the cover 981. When the cover 981 is in the obscuring orientation, the fasteners 998 may be closed to close the upper opening 996 and secure the cover 981 in the

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obscuring orientation. To transition the cover 981 to a viewing orientation, a portion of the cover 981 may be displaced upwardly (e.g., from the bottom edge such as by the drainage tube 912) to reveal at least a lower portion of the fluid compartment 921. Alternatively, the fasteners 998 may be opened to open the upper opening 996 and a portion of the cover 981 may be displaced downwardly to reveal at least an upper portion of the fluid compartment 921.

As shown in FIG. 25, in certain embodiments, a bodily fluid drainage assembly 1000 can include a drainage bag assembly 1010 comprising a cover 1081 that can be formed from or otherwise include an opaque material (or at least partially opaque material capable of obscuring the appearance of a fluid within the drainage bag assembly 1010 when in use). For example, the cover 1081 may be formed from polyvinyl chloride, polyurethane, vinyl, polymeric, or any other suitable material. The cover 981 may comprise front and back panels that can obscure fluid compartments 1021, 1022 when in use. The front and back panels of the cover 1081 may be configured to obscure substantially all of the fluid compartments 1021, 1022. As depicted in FIG. 25, a top edge of the front panel of the cover 1081 is coupled (e.g., welded) to a top edge of the front fluid compartment 1021. The weld extends substantially across a width of the fluid compartment 1021 such that the cover 1081 obscures the fluid compartment 1021 from top to bottom and from side to side. In other words, the cover 1081 is coupled to the fluid compartment 1021 adjacent an elongate member 1051 of the hanger 1050. In some embodiments, the weld may follow a contour of the elongate member 1051. The fluid compartment 1022 and back panel of the cover 1081 can be configured likewise.

The illustrated embodiment of FIG. 25 depicts an inlet port 1016 in fluid communication with the fluid compartment 1022 of the drainage bag assembly 1010. Alternatively, the fluid port 1016 may be in fluid communication with the fluid compartment 1021. An elbow connector 1094 is rotationally coupled to and extends upward from the inlet port 1016 at one end and is fluidly coupled to a drainage tube 1012 at an opposite end. The elbow connector 1094 is configured to be rotated relative to the inlet port 1016 to facilitate redirecting the drainage tube 1012 (e.g., such that the drainage tube 1012 can be oriented at any direction in relation to the assembly 1010). The elbow connector 1094 can include a fixed angle bend ranging from about 30 degrees to about 60 degrees, from about 40 degrees to about 50 degrees, and can be about 45 degrees relative to a vertical axis of the elbow connector 1094. In other embodiments, the elbow connector 1094 may include a swivel joint disposed at the bend of the elbow connector 1094. The swivel joint may allow the angle of the elbow connector 1094 to be adjustable from about zero degrees to about 90 degrees, from about 30 degrees to about 60 degrees, and from about 40 degrees to about 50 degrees relative to the vertical axis of the elbow connector 1094.

A rotation tab or wing 1098 may be coupled to the elbow connector 1094. The rotation tab 1098 may extend outwardly from the vertical axis and be disposed adjacent the bend of the elbow connector 1094. The user may grasp the rotation tab 1098 to rotate the elbow connector 1094 to change an orientation of the drainage tube 1012. In other embodiments, the elbow connector 1094 may not include the rotation tab 1098 and the elbow connector 1094 may be rotated by grasping a body of the elbow connector 1094. An anti-rotation handle 1095 may also optionally be fixedly coupled to an upper portion of the inlet port 1016. The anti-rotation handle 1095 may include at least one lateral

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extension configured to be gripped by a user to rotationally secure the inlet port **1016** relative to the fluid compartment **1022**. When grasped by a user, the anti-rotation handle **1095** can prevent rotation of the inlet port **1016** as the elbow connector **1094** is rotated. If the inlet port **1016** is rotated, the material of the fluid compartment **1022** may tear or rip and compromise the fluid integrity of the fluid compartment **1022**. The anti-rotation handle **1095** may be used in cooperation with the rotation tab **1098** to rotate the elbow connector **1094** to re-orient the drainage tube **1012** without compromising the fluid integrity of the fluid compartment **1022**. In other embodiments, no anti-rotation handle **1095** is used.

FIG. **25** illustrates the drainage bag assembly **1010** including a retention band **1093**. The band **1093** may be fixedly or releasably coupled to side edges of the fluid compartments **1021**, **1022**. The band **1093** may be formed from the same material used to form the fluid compartments **1021**, **1022**, or a different material. The band **1093** may be about one inch wide and have a length that accommodates expansion of the fluid compartments **1021**, **1022** when filled with fluid. The band **1093** may be configured to retain an outlet port **1013** in an upward vertical orientation when the fluid compartments **1021**, **1022** are not being drained. A distal portion of an output regulator **1014** may be selectively disposed inside of the band **1093** to retain the outlet port **1013** in an upward vertical orientation. The band **1093** can also be configured to be lifted and/or stretched upwards such that the outlet port **1013** can be transitioned from the upward retained position to a position where it extends outwardly and/or downwardly for drainage. In other embodiments, the band **1093** may include portions coupled to and extending from each of the fluid compartments, **1021**, **1022**. The portions may be selectively coupled using any suitable fastener, such as a button, a snap, adhesive, hook-and-loop material, etc.

Any methods disclosed herein comprise one or more steps or actions for performing the described method. The method steps and/or actions may be interchanged with one another. In other words, unless a specific order of steps or actions is required for proper operation of the embodiment, the order and/or use of specific steps and/or actions may be modified.

References to approximations are made throughout this specification, such as by use of the term “substantially.” For each such reference, it is to be understood that, in some embodiments, the value, feature, or characteristic may be specified without approximation. For example, where qualifiers such as “about” and “substantially” are used, these terms include within their scope the qualified words in the absence of their qualifiers. For example, where the term “substantially perpendicular” is recited with respect to a feature, it is understood that in further embodiments, the feature can have a precisely perpendicular configuration.

Similarly, in the above description of embodiments, various features are sometimes grouped together in a single embodiment, figure, or description thereof for the purpose of streamlining the disclosure. This method of disclosure, however, is not to be interpreted as reflecting an intention that any claim require more features than those expressly recited in that claim. Rather, as the following claims reflect, inventive aspects lie in a combination of fewer than all features of any single foregoing disclosed embodiment.

The claims following this written disclosure are hereby expressly incorporated into the present written disclosure, with each claim standing on its own as a separate embodiment. This disclosure includes all permutations of the independent claims with their dependent claims. Moreover,

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additional embodiments capable of derivation from the independent and dependent claims that follow are also expressly incorporated into the present written description.

Without further elaboration, it is believed that one skilled in the art can use the preceding description to utilize the invention to its fullest extent. The claims and embodiments disclosed herein are to be construed as merely illustrative and exemplary, and not a limitation of the scope of the present disclosure in any way. It will be apparent to those having ordinary skill in the art, with the aid of the present disclosure, that changes may be made to the details of the above-described embodiments without departing from the underlying principles of the disclosure herein. In other words, various modifications and improvements of the embodiments specifically disclosed in the description above are within the scope of the appended claims. Moreover, the order of the steps or actions of the methods disclosed herein may be changed by those skilled in the art without departing from the scope of the present disclosure. In other words, unless a specific order of steps or actions is required for proper operation of the embodiment, the order or use of specific steps or actions may be modified. The scope of the invention is therefore defined by the following claims and their equivalents.

The invention claimed is:

1. A bodily fluid drainage assembly, comprising:

a first fluid compartment comprising a first elongate member;

a second fluid compartment comprising a second elongate member, the second fluid compartment being in fluid communication with the first fluid compartment, wherein the first and second fluid compartments are configured to be parallel to one another when in an assembled configuration;

a base extending between the first elongate member and the second elongate member such the first elongate member and the second elongate member are spaced apart from one another in the assembled configuration; and

a hanger configured to couple to the base to suspend the bodily fluid drainage assembly from a support, wherein the base comprises a slot for coupling the hanger to the base.

2. The bodily fluid drainage assembly of claim 1, wherein the base comprises a first base portion and a second base portion that are coupled together in the assembled configuration.

3. The bodily fluid drainage assembly of claim 2, wherein the slot is defined by the first base portion and the second base portion.

4. The bodily fluid drainage assembly of claim 3, wherein the hanger is inserted into the slot and turned to lock a first base portion and a second base portion together in the assembled configuration.

5. The bodily fluid drainage assembly of claim 1, wherein the first and second fluid compartments are defined by one or more seams.

6. The bodily fluid drainage assembly of claim 1, further comprising:

a first cover coupled to the first fluid compartment and configured to obscure observation of at least a portion of a bodily fluid through a first panel of the first fluid compartment when the bodily fluid is retained within the first fluid compartment.

7. The bodily fluid drainage assembly of claim 6, wherein the first cover comprises at least one volume indicator.

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8. The bodily fluid drainage assembly of claim 6, wherein the first cover is configured such that at least a portion of the first cover is removable from the bodily fluid drainage assembly.

9. The bodily fluid drainage assembly of claim 6, wherein the first cover is coupled to the bodily fluid drainage assembly at an upper end of the first fluid compartment, and wherein a lower end of the first cover can be displaced so as to transition the first cover from the obscuring orientation to a viewing orientation.

10. The bodily fluid drainage assembly of claim 1, wherein the first fluid compartment further comprises:
an inlet port configured to couple to a drainage tube and configured to receive drainage fluid from the drainage tube.

11. The bodily fluid drainage assembly of claim 1, further comprising an outlet tube, wherein the outlet tube is disposed on an outlet tube section that is configured to extend outwardly from the first and second fluid compartments, wherein the outlet tube is configured to be disposed in an upright configuration when the outlet tube section is folded inwardly, and wherein the outlet tube is disposed in a downward configuration when the outlet tube section is extended outward.

12. A bodily fluid drainage assembly, comprising:

a first fluid compartment comprising a first elongate member;

a second fluid compartment comprising a second elongate member, the second fluid compartment being in fluid communication with the first fluid compartment, wherein the first and second fluid compartments are configured to be parallel to one another when in an assembled configuration;

a base extending between the first elongate member and the second elongate member such the first elongate member and the second elongate member are spaced apart from one another in the assembled configuration;

a hanger configured to couple to the base to suspend the bodily fluid drainage assembly from a support, wherein the base further comprises a slot for coupling the hanger to the base; and

a first cover configured to obscure observation of at least a portion of a bodily fluid through a first panel of the first fluid compartment when the bodily fluid is retained within the first fluid compartment.

13. The bodily fluid drainage assembly of claim 12, wherein the base comprises a first base portion and a second base portion that are coupled together in the assembled configuration.

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14. The bodily fluid drainage assembly of claim 13, wherein the slot is defined by the first base portion and the second base portion, and wherein the hanger is inserted into the slot and turned to lock the first base portion and second base portion together in the assembled configuration.

15. The bodily fluid drainage assembly of claim 12, wherein the first and second fluid compartments are defined by one or more seams.

16. The bodily fluid drainage assembly of claim 12, further comprising:

a second cover coupled to the second fluid compartment and configured to obscure observation of at least a portion of the bodily fluid through a second panel of the second fluid compartment when the bodily fluid is retained within the second fluid compartment.

17. The bodily fluid drainage assembly of claim 12, wherein the first cover comprises at least one volume indicator.

18. A method of suspending a drainage system from a hospital bed rail, comprising:

obtaining a drainage system, wherein the drainage system comprises:

a first fluid compartment comprising a first elongate member;

a second fluid compartment comprising a second elongate member, the second fluid compartment being in fluid communication with the first fluid compartment, wherein the first and second fluid compartments are configured to be parallel to one another when in an assembled configuration;

a base extending between the first elongate member and the second elongate member such the first elongate member and the second elongate member are spaced apart from one another in the assembled configuration, wherein the base comprises a slot; and

a hanger configured to couple to the base to suspend the drainage system from a support;

inserting the hanger into the slot in the base; and
coupling the hanger to the hospital bed rail.

19. The method of claim 18, further comprising:

draining bodily fluid from the drainage system, wherein draining comprises changing an outlet tube from an upright configuration to a downward configuration.

20. The method of claim 18, wherein the base comprises a first base portion and a second base portion that are coupled together in the assembled configuration.

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